High-Resolution Modeling of Compound Urban Flooding: 2. Flood Driver Analysis

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Abstract

Compound flooding in coastal urban environments arises from the interaction among precipitation, upstream inflow, and coastal water levels, yet whether these drivers combine linearly or through non-linear threshold dynamics, and where each regime prevails, remain unresolved. This study develops a physically interpretable framework to quantify driver interactions in Savannah, Georgia, by coupling a multi-scale hydrodynamic system (1D–2D HEC-RAS integrated with PCSWMM) with both linear (Partial Least Squares Regression, PLSR) and non-linear (Random Forest, XGBoost) surrogate models trained on a 32-scenario Latin Hypercube Sampling ensemble. Learning curve analysis confirms that model performance plateaus before the full sample size is reached, validating the surrogate approach. Results show that compound urban flooding is fundamentally non-linear: cross-validated PLSR yields persistently negative R^2 for urban, ditch, and water reservoir surface types regardless of ensemble size, while Random Forest achieves R^2 exceeding 0.64 at these nodes, demonstrating structural model failure rather than a sample-size artifact. In addition, driver dominance is spatially stratified: downstream surge controls FEMA Zone AE (median importance > 0.8), while precipitation governs Zone X via a saturation threshold at approximately 1.2 times baseline intensity, with upstream inflow negligible for peak depth throughout, though it exhibits a non-monotonic influence on flood duration via upstream wetland storage dynamics. Within Zone X, where 66% of repetitive loss areas are located, non-linear precipitation thresholds are the primary driver, exposing a blind spot in single-driver regulatory mapping. Furthermore, flood residual time reveals a previously uncharacterized predictability regime: channel and auxiliary ditch nodes are structurally unpredictable by any model because scalar boundary conditions lack the temporal forcing information (storm duration, surge duration) required to resolve recession dynamics. These findings demonstrate that compound flood risk assessment requires non-linear, spatially resolved frameworks and define specific data requirements for flood duration forecasting.

1. Introduction

Compound flooding in coastal urban areas arises from the concurrent or sequential interaction of multiple drivers, including storm surge, riverine overflow, and intense precipitation, with combined impacts frequently exceeding the sum of their individual effects (Sebastian, 2022; Xu et al., 2022). These hazards are intensifying in low-lying coastal and deltaic regions under sea-level rise, urbanization, and climate change (Lin et al., 2016; Nicholls, 2002; Taherkhani et al., 2020; Vitousek et al., 2017). In the southeastern United States, where flood-related damages exceeded $3.5 billion in 2023 alone, accurately characterizing the relative contribution of each driver and their interactions is essential for targeted risk management (NOAA, 2024; Zscheischler et al., 2018).

This study is part of a two-paper investigation of compound flood risk in coastal urban environments. In Part 1 (Kim et al., 2026), we developed and validated a high-resolution, multi-scale hydrodynamic model for Savannah, Georgia, integrating SWMM/PCSWMM for urban drainage with HEC-RAS for riverine dynamics. That model provides the deterministic simulation platform used here. In this companion paper (Part 2), we address a question that high-resolution simulation alone cannot answer: which flood driver or combination of drivers controls the observed flood response at each location, and whether those controls are linear or fundamentally non-linear.

Answering this question requires an ensemble of scenarios that spans the multi-dimensional driver space, a requirement that purely observational or single-event studies cannot meet (Schmidt et al., 2020; Jin Teng et al., 2017). While high-resolution hydrodynamic models can simulate individual events in detail, their computational cost precludes exhaustive exploration of driver combinations at fine resolution (Néelz, 2009; J. Teng et al., 2017). To overcome these limitations and account for uncertainty in future climate and land use, we apply LHS to generate a broad ensemble of boundary condition scenarios (Helton & Davis, 2003; Jung et al., 2011).

A central, unresolved question in compound flood research is whether the combined impact of multiple drivers is merely additive (linear) or governed by threshold-dependent, non-linear interactions—and, critically, whether the answer varies spatially within a single catchment. To resolve this, we introduce a dual-modeling diagnostic framework. Partial Least Squares Regression (PLSR) treats the null hypothesis of linear driver superposition as the baseline, while Random

Forest (RF) and XGBoost capture non-linear threshold responses. By comparing their cross-validated performance at every computational node, we quantify the linearity gap—the degree to which non-linear models outperform the linear baseline—and map its spatial distribution across land-use types and regulatory flood zones.

Applied to Savannah, Georgia, a coastal city where 66% of repetitive flood loss areas fall outside the designated 1% floodplain, this framework yields three principal findings: (1) compound urban flooding is fundamentally non-linear at the node scale, as demonstrated by persistently negative cross-validated R^2 for PLSR at urban and drainage nodes regardless of ensemble size; (2) flood driver dominance is spatially stratified along a coherent hydraulic gradient from surge-controlled coastal zones to precipitation-controlled inland areas, exposing a systematic blind spot in single-driver basis mapping; and (3) flood residual time reveals a previously uncharacterized 'structural unpredictability' regime at channel nodes, arising from the absence of temporal forcing information in scalar boundary conditions.

2. Research Domain and Hydrodynamic Modeling

2.1 Research Domain

The study area focuses on the City of Savannah, Georgia, situated along the Savannah River, and is illustrated as an urban model in Fig. 1(A). Characterized by a dense network of canals and tributaries that discharge into the Atlantic Ocean, Savannah is uniquely vulnerable to coastal flooding due to its low elevation, proximity to tidal influences, and interconnected hydrologic systems. The city's stormwater infrastructure, comprising more than 23,000 outfalls, plays a critical role in managing runoff but is increasingly stressed by heavy precipitation, backwater effects from riverine surges, and tidal fluctuations. As one of Georgia's fastest-growing metropolitan areas, Savannah also faces compounding challenges from aging infrastructure, declining infiltration capacity due to urbanization, and exposure to rising sea levels along the southeastern U.S. coast.

Historical records at Fort Pulaski, near the Savannah River estuary, reveal a dramatic increase in coastal flooding events, from just 0.15 per year between 1950 and 1969 to 7.5 per year from 2011 to 2023, demonstrating the intensifying impacts of sea-level rise and climate change (NOAA, 2024; Ohenhen et al., 2024; Parker et al., 2023; Sweet et al., 2021). Fig. 1(B) illustrates this trend, showing a sharp rise in flood frequencies at key Atlantic coastal sites. The Fifth National Climate

Assessment further projects a 45% increase in extreme precipitation events across southeastern U.S. counties under a 2°C global warming scenario, highlighting additional fluvial and pluvial stressors (Crimmins et al., 2023).

Notably, recent data from OpenFEMA reveal that flood damage is disproportionately occurring in areas designated as lower-risk (0.2% annual exceedance probability zones), with over 66% of repetitive-loss areas and 65% of previously flooded structures located outside the 1% floodplain. This trend suggests that traditional flood risk delineations may underrepresent actual vulnerabilities in urban coastal areas, particularly under compound flood scenarios.

To investigate these dynamics, Hurricane Matthew (2016) is selected as the representative case event due to its substantial and well-documented impacts on the Savannah region. Among hurricanes affecting the southeastern coast between 2008 and 2023, Matthew produced one of the most extreme combinations of precipitation and storm surge, making it an ideal baseline for analyzing compound flooding mechanisms in this high-risk coastal environment.

2.2 Hydrodynamic Modeling

For completeness, we briefly summarize the hydrodynamic model developed for the study area; full technical details are provided in a companion paper (Kim et al., 2026). The integrated framework consists of a riverine flood model for the Savannah River Basin (Fig. 2A) and an urban flood model focusing on Savannah and Garden City, GA (Fig. 2B). HEC-RAS was selected for the riverine component due to its proven capability to simulate compound flooding under variable boundary conditions, including river discharge, tidal fluctuations, and precipitation. This model supports flexible spatial resolution to accommodate the region's complex topography and hydraulic structures. Key inputs include hydro-enforced digital elevation models (DEMs), stormwater infrastructure data in GIS format, and high-resolution hydrologic datasets from NHDPlus HR (Moore et al., 2019), which together provide a detailed representation of the hydrologic and physical landscape.

The urban flood model is created using SWMM/PCSWMM and features a hierarchical, multi-scale structure to simulate both surface runoff and subsurface drainage. It includes about 23,000 stormwater network components, such as underground pipelines, culverts, weirs, and open channels. Grid resolutions vary from 10 to 30 meters, depending on land use, slope, and hydraulic complexity. This adaptive resolution method ensures that high-risk areas are modeled with greater detail while keeping computational efficiency across the entire region.

Simulations proceed in a sequential, nested manner: a coarse-resolution model (30–50 m) initially captures regional flood behavior and supplies boundary conditions—such as water levels and flow rates—for more detailed models (10–20 m) that focus on localized urban flooding. This step-by-step approach effectively models the interaction between riverine and urban flooding processes, which is crucial for evaluating compound flood risks in coastal areas.

Model validation uses multiple independent data sources. USGS gauge stations provide time-series river stage data, while FEMA and GEMA records supply flood extent and damage information. Supplementary validation is performed using media reports about urban flood impacts. For areas without gauges, Synthetic Aperture Radar (SAR) imagery detects flood extent, providing valuable spatial validation where ground-based data are missing. A single simulation of the fluvial model for a 14-day flood event took 11 minutes, whereas the urban flood model for 4 days took 9 hours per scenario.

3 Flood Driver Analysis Methods

We conducted a flood driver analysis by categorizing the key contributing factors—precipitation, sea-level rise, storm surge, urbanization, and aging stormwater infrastructure—into three primary boundary condition variables: precipitation amount (X_3), upstream conditions (X_1, representing integrated basin discharge and reservoir management), and downstream conditions (X_2, representing coastal water levels including sea-level rise and storm surge). Using Hurricane Matthew (2016) as a baseline, we systematically varied combinations of these variables to evaluate their effects on two flood-response metrics: (1) maximum water depth, the peak depth recorded per grid cell across scenarios, and (2) flood residual time, the duration during which surface water depth remains above 5 cm. To rigorously demonstrate that compound urban flooding involves physically non-linear interactions among these drivers—rather than merely additive superposition—we employ a three-stage proof framework. First, quantitative performance benchmarking (R^2 comparison between PLSR and RF/XGBoost) provides existence proof: if non-linear models substantially outperform a linear baseline, the physical system is demonstrably non-linear. Second, Partial Dependence Plots provide mechanistic evidence by isolating the marginal response of each metric to each driver, identifying specific physical thresholds. Third, spatial feature importance mapping provides regime proof by demonstrating that non-linearity is spatially heterogeneous, with distinct hydraulic regimes partitioning the domain.

3.1 Linear approach using Partial Least Squares Regression (PLSR) Analysis

To analyze the relationship between input drivers and flood responses with a linear approach, we employed Partial Least Squares Regression (PLSR), a robust multivariate statistical technique commonly used in disciplines such as chemometrics, genomics, and environmental modeling (Groen & Kapetanios, 2016; Martens & Martens, 2001; Nguyen & Rocke, 2002; Wold et al., 1984). PLSR integrates principles from principal component analysis (PCA) and multiple regression to identify latent structures that best explain variation in the response variables. All variables were normalized to assess relative rather than absolute effects, allowing identification of dominant patterns irrespective of scale. Additionally, we applied the Partial Rank Correlation Coefficient (PRCC) method to determine the relative importance of each flood driver. This dual approach provides complementary insights into both the magnitude and rank order of influence, enhancing our understanding of the drivers of compound flooding in the study area.

PLSR is a multivariate statistical technique that projects predictor and response variables into a new space, maximizing their covariance to identify the most relevant latent components for predicting the response variable. In this study, the predictor variables $\mathbf{X} \in \mathbf{R}^{n \times p}$ include precipitation amount, upstream conditions, and downstream conditions, while the response variable $\mathbf{Y} \in \mathbf{R}^{n \times q}$ represents water stage characteristics (e.g., water surface elevation and surface water depth). n, p, and q represent the number of observations or scenarios, the number of predictor variables, and the number of response variables (e.g., water surface elevation), respectively.

The PLSR model decomposes $\mathbf{X}$ and $\mathbf{Y}$ into score and loading matrices that maximize their covariance (Wold et al., 2001). The regression relationship is:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{F} \tag{1}$$

where $\mathbf{B} = \mathbf{W}\left(\mathbf{P}^{\mathrm{T}}\mathbf{W}\right)^{-1}\mathbf{Q}^{\mathrm{T}}$ is the regression coefficient matrix, $\mathbf{W}$ contains the weight vectors maximizing covariance between $\mathbf{X}$ and $\mathbf{Y}$, and $\mathbf{P}$ and $\mathbf{Q}$ are the loading matrices for predictors and responses, respectively. The number of latent components was determined by leave-one-out cross-validation. The predicted response matrix $\hat{\mathbf{Y}} = \mathbf{XB}$ yields flood characteristics including maximum surface water depth and residual time. Full algorithmic details follow Wold et al. (2001). This methodology enables a comprehensive analysis of how various flood drivers impact flood characteristics, providing insights essential for urban flood management and resilience planning.

To complement the PLSR analysis, Partial Rank Correlation Coefficients (PRCC) were computed to assess the rank-order influence of each driver on the response variable while controlling for interdependencies among inputs (Saltelli et al., 2004). PRCC isolates each predictor's contribution by correlating the residuals of rank-transformed regressions of both the predictor and the response on all remaining predictors. Values near ±1 indicate strong monotonic influence after removing confounding effects. Detailed derivations and equations are included in a supplement file.

3.2 Nonlinear approach using Machine Learning-Based Analysis

Tree-based ensemble algorithms, including Random Forest (RF) and Extreme Gradient Boosting (XGBoost), were selected for their ability to capture the threshold-dependent, non-linear dynamics that characterize urban flood response (Abedi et al., 2022; Ma et al., 2021; Wang et al., 2023). Unlike linear approaches, tree-based models recursively partition the input space to isolate sharp physical thresholds, such as stormwater pipe capacity exceedance. We refer to these collectively as infrastructure threshold effects, including stormwater pipe capacity exceedance and backwater-induced surcharge (Chen et al., 2025). Critically for flood risk management, these models integrate natively with explainable AI (XAI) frameworks, feature importance, partial dependence, and SHAP, enabling the statistical predictions to be systematically translated into physical hydrodynamic insights (Erion et al., 2021; Rudin, 2019).

3.2.1 Random Forest

Random Forest (RF) constructs an ensemble of B independently trained decision trees on bootstrap samples, with predictions averaged across trees (Breiman, 2001).

$$\grave{f}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x) \tag{2}$$

At each node, a random subset of $m_{try} < p$ is evaluated, where p is the total number of input features as predictors, and m_{try} is the number of features randomly sampled at each split node. The split minimizing within-node variance is selected. This averaging mechanism acts as a lowpass filter for variance, allowing the model to capture sharp discontinuities, such as threshold-driven overtopping, while maintaining stability.

To generate unbiased error metrics without sacrificing training data for a holdout set, Out-of-Bag (OOB) predictions were utilized. For each tree, approximately one-third of the data is left unseen during training. The OOB prediction for a given node is calculated by averaging predictions only from trees that excluded that sample from their bootstrap dataset. This provides a computationally efficient, internally cross-validated estimate of generalization error (R^2 and RMSE) within the 32-scenario ensemble.

3.2.2 XGBoost

XGBoost constructs an ensemble of *K* regression trees sequentially via gradient boosting, where each new tree corrects the residual error of the preceding ensemble (Chen & Guestrin, 2016). The objective function combines a convex loss term with L2 regularization of tree complexity — penalizing both the number of leaves, T, and the magnitude of leaf weights, *w* — to control overfitting. This gradient-based optimization allows XGBoost to learn subtle non-linear interactions between drivers that linear models and standard bagging approaches may miss.

Because sequential boosting is more susceptible to overfitting than bagging, a strict 5-fold cross-validation strategy was applied. The data was partitioned into five equally sized folds; the model was iteratively trained on four folds and tested on the held-out fifth, rotating until an out-of-sample prediction was generated for every instance. The aggregate of these CV predictions was used to calculate final performance metrics, ensuring that reported accuracy reflects predictive capability on held-out scenarios.

Feature importance was quantified using the Gain metric.

$$Gain_s = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{\left(G_L^2+G_R^2\right)}{H_L+H_R+\lambda}\right] - \gamma \tag{3}$$

where G, $H$are first and second-order gradient statistics, λ is L2 regularization, and γ is the minimum gain threshold. For a given feature *F*, importance is the cumulative reduction in the loss function across all splits on *F* in all trees:

$$Importance(F) = \sum_{t=1}^{K} \sum_{s \in t} 1\,[v_s = F] \cdot Gain \tag{4}$$

where K is the total number of trees, s indexes all splits across tree t, v_s is the feature used at split s, and $Gain$ is the reduction in the loss function achieved at split s derived from the gradient boosting structure (Chen et al., 2019). This metric enabled spatially distributed, physics-informed interpretation of driver dominance.

3.2.3 Model Selection Criteria

RF reduces variance by averaging and is well-suited to continuous response variables such as flood depth. XGBoost reduces bias through sequential learning and is advantageous for zero-inflated distributions such as residual time. Model selection between the two is guided by (1) response variable properties (zero-inflation, skewness) and (2) whether a substantial improvement ($\Delta R^2 > 0.15$) over the alternative is achieved. Empirical comparison follows in Sections 4.3.1–4.3.2 and 4.3.5.

3.3 Variable Sampling

Using Hurricane Matthew (2016) as a baseline event, boundary conditions were systematically varied using Latin Hypercube Sampling (LHS) to ensure efficient coverage of the three-dimensional input space (Helton & Davis, 2003). The three input variables are upstream conditions (X_1, representing integrated basin discharge and reservoir management), downstream conditions (X_2, representing coastal water levels including sea-level rise and storm surge), and precipitation amount (X_3). LHS divides each variable's range into N equal-probability intervals and samples once per interval, reducing output variance relative to random sampling while avoiding the combinatorial cost of full factorial designs.

Downstream and upstream conditions were modified by specific increments, and precipitation was adjusted as a percentage relative to the Matthew baseline, yielding a total of 32 scenarios. The fluvial model produced 1D water stage at 925 river nodes and 2D raster time series at 30 m resolution; the urban model simulated 32 scenarios, generating 10,720 computational nodes per scenario. Each node was analyzed independently to characterize spatial heterogeneity in driver importance; the underlying hydrodynamic model accounts for physical connectivity between nodes. We empirically verify sample adequacy through learning-curve analysis in Sections 4.3.1 and 4.3.2, demonstrating that model performance plateaus before the full sample size is reached for most surface types.

Because LHS varies independently, the ensemble does not represent the positive correlation between peak storm surge and precipitation totals observed during landfalling Atlantic hurricanes. While LHS decorrelates the input space to isolate marginal driver sensitivities (first-order attribution), the limitations of omitting joint probability structures are addressed in Section 5. The importance scores derived in Section 4 therefore characterize marginal driver sensitivities under a decorrelated input space and should be interpreted as first-order attribution results. Joint probability sampling using multivariate copulas, conditioned on the regional climatological dependence structure, is identified as a priority extension in Section 5.4.

4. Results

4.1 Baseline Model and Non-linearity Problem

The fluvial flood model was validated against real-world flood data from major hurricanes, including Matthew (2016), Irma (2017), and Idalia (2023), as well as through the PIGA approach (Kim et al., 2024). The 2D urban flood model was validated using Hurricane Matthew, with reference to local media reports and Sentinel-1 SAR imagery (Kim et al., 2026). Regarding fluvial modeling, 32 scenarios were generated, with each scenario represented by a time series for each node (river station in HEC-RAS), treated as a data point. River stage datasets generated by the fluvial model were used as boundary conditions for the urban areas.

Fig. 3 presents the baseline model results for Hurricane Matthew. The outputs capture the temporal evolution of inundated areas, maximum water depth, and flood retention (residual) time, along with their statistical characteristics (boxplots and histograms). The spatial distribution of maximum water depth shows that deeper flooding primarily occurs along river channels connected to the coast, including the Savannah River, Vernon River, Moon River, and Wilmington River, which are strongly influenced by tides and storm surge. In contrast, downtown areas experience relatively shallow flooding, largely due to higher elevations and effective drainage systems. The histograms indicate non-symmetric distributions for inundated areas, maximum depth, and residual time. Notably, the boxplots for maximum depth and residual time show heavy right tails, suggesting that certain areas experienced disproportionately high water levels and prolonged inundation. Importantly, the heavy right-tailed distributions of maximum depth and residual time (Figs. 3E and 3I) provide an initial empirical indicator of system non-linearity. A strictly linear flood system driven by near-uniform LHS inputs would produce approximately symmetric output distributions. The pronounced positive skewness — where a small subset of nodes experiences disproportionately amplified responses — is consistent with threshold-governed non-linear dynamics in portions of the drainage network, motivating the rigorous analysis in Sections 4.3.1 and 4.3.2.

4.2 Fluvial Flood Driver Analysis

The fluvial flood model provides boundary conditions for the urban flood model along the riverine system. A flood-driver analysis was conducted for the 1D channels, as shown in Fig. 4. In Fig. 4A, the PLSR effectively projects the maximum surface water elevation (MSWE), producing results that closely match the fluvial model outputs across all scenarios. Fig. 4B illustrates the difference in MSWE between the PLSR and the fluvial model, which follows a near-normal distribution with a mean of zero and minimal variance, confirming the accuracy of PLSR in quantifying water levels in the Savannah River. These results are consistent with the previously developed physics-informed geostatistical approach (PIGA), which demonstrated that river water levels can be represented by a linear system model based on spatial correlations. Building on that, this study shows that MSWE can also be expressed as a linear function of key flood drivers—precipitation, upstream inflows, and downstream (coastal) conditions—validating the robustness of PLSR for analyzing flood dynamics in the Savannah River Basin.

Fig. 4C–4E present the spatial distributions of the PLSR weights for downstream conditions, precipitation, and upstream conditions, respectively. Downstream changes exert the strongest influence on MSWE, particularly in coastal and near-coastal areas, with their effect gradually weakening upstream (Fig. 4C). This dominance is further supported by the histogram of coefficients (Fig. 4F), which is centered on high values. Precipitation (Fig. 4D) exerts a broader but less intense effect, with notable influence around urban areas due to runoff and drainage. Its coefficient histogram (Fig. 4G) is more evenly distributed, suggesting a consistent but moderate contribution across the domain. Upstream conditions (Fig. 4E) primarily affect headwater regions, with diminishing influence toward the coast; their histogram (Fig. 4H) is centered on lower values, reflecting a localized impact.

The high R^2 values confirm a strong linear relationship between flood drivers and MSWE, emphasizing that compound flooding results from interacting processes rather than a single dominant factor. This linearity is physically consistent with the hydraulic characteristics of the lower Savannah River. The channel's wide, prismatic geometry and low bed gradient maintain gradually varied subcritical flow conditions across the full range of hurricane-scale discharges examined, keeping stage-discharge relationships within the approximately linear portion of Manning's resistance curve. Furthermore, the dominance of the downstream tidal prism over upstream riverine discharge in the lower reach means that MSWE is primarily governed by a single, monotonically varying boundary forcing — a configuration inherently conducive to linear representation. The spatial patterns further reveal that coastal influences can propagate inland, with sea-level rise (SLR) projected to affect flood risks up to 35 km from Fort Pulaski. These findings highlight the importance of integrating multiple drivers, including coastal forcing, precipitation, and upstream hydrology, into flood risk assessments, particularly for urbanized coastal regions.

4.3 Urban Flood Driver Analysis

The fluvial analysis in Section 4.2 demonstrated that riverine water-surface elevation responds approximately linearly to the three boundary condition drivers. This section shows that the same linearity holds for domain-integrated flood metrics (total inundated area) but breaks down fundamentally at the node level, where threshold-governed urban hydraulics produce responses that linear models are structurally incapable of representing. This scale-dependent transition from linearity to non-linearity is the central finding of this study and has direct implications for which modeling frameworks are appropriate for different components of compound flood risk assessment.

A methodological caveat applies to all important results that follow. Because LHS varies each driver independently, the ensemble does not reproduce the positive correlation between peak storm surge and precipitation totals observed during landfalling Atlantic hurricanes. The driver importance scores therefore represent marginal sensitivities under a decorrelated input space—first-order attributions that identify which driver each node responds to most strongly, but not the synergistic amplification that occurs when correlated extremes co-occur (see Section 5.4).

Across all scenarios, the flood model assessed the effects of various flood drivers on urban flooding, focusing on maximum water depth and residual flood duration. Fig. 5 presents representative simulation results under selected scenarios. Fig. 5A–5C show the downstream boundary conditions, upstream boundary conditions, and mean precipitation, respectively. Fig. 5D–5F then compare their relative impacts on total inundated area, peak water depth during the hurricane, and the histogram of residual time for inundated areas.

At the catchment scale, flood response remains approximately linear: PLSR achieves $R^2 = 0.93$–0.99 for total inundated area across all fixed-driver conditions (Table S1), with downstream conditions and precipitation acting as co-dominant drivers and upstream inflow contributing negligibly (PLSR coefficient < 0.05; PRCC = 0.003). This catchment-scale linearity is consistent with the fluvial analysis in Section 4.2. However, spatial integration structurally obscures the nonlinear threshold dynamics governing flooding at individual nodes. The node-level analysis that follows demonstrates that urban flood analysis is fundamentally incompatible with linear representation—requiring non-linear models for accurate characterization. Specifically, the urban flood threshold effect determines peak water depth and inundation duration at specific locations. The following analysis of maximum water depth and flood residual time demonstrates that these point-scale metrics are fundamentally non-linear, requiring machine learning-based approaches for accurate representation, and that the spatial distribution of this non-linearity defines distinct hydraulic regimes with direct implications for flood risk management.

Before examining point-scale flood metrics, the global partial dependence analysis (Fig. 6) characterizes the domain-wide marginal response of both maximum depth and flood residual time to each boundary condition driver. Three driver-response regimes are evident. Downstream conditions produce a monotonic increase in both metrics, with a proportionally larger marginal range for duration (~1.9 hours) than for depth (~0.38 m), reflecting the sustained influence of tidal forcing on drainage recovery. Precipitation exhibits a non-linear saturation threshold at approximately 1.0× baseline intensity for both metrics, consistent with drainage network surcharge beyond which additional rainfall yields diminishing marginal returns. Upstream water stage exerts negligible influence on peak depth (range < 0.02 m) and a weakly decreasing effect on duration, consistent with upstream wetland engagement attenuating the surge. These domain-wide trends provide the context for the node-level analyses that follow, which demonstrate that the smooth aggregate behavior conceals threshold-governed non-linearities at individual computational nodes.

4.3.1 Maximum Water Depth Analysis

To rigorously quantify the complexity of compound flooding, we compared the predictive performance of a linear baseline model (Partial Least Squares Regression, PLSR) with that of a non-linear machine learning model (Random Forest, RF). This comparison serves as a proxy for the system's physical complexity: where the linear model fails, the underlying flood physics are dominated by non-linear interactions. Figure 7 presents the performance of RF projection. While data-driven models typically require large datasets, the Random Forest models in this study were trained on 32 hydro-climatological scenarios per node. This sample size is sufficient because the models emulate a deterministic physical system, yielding a high signal-to-noise ratio. Furthermore, the feature space is strictly limited to three boundary-condition drivers, thereby avoiding the curse of dimensionality. To maximize data utilization and prevent overfitting, model validation was conducted exclusively using Out-of-Bag (OOB) error estimates, eliminating the need for a separate hold-out testing set.

Fig. 7 presents the Random Forest regression performance for maximum water depth prediction at both the global domain scale and for Urban nodes in isolation. At the global level (Fig. 7-B), the RF model achieves an MAE of 0.026 m and RMSE of 0.059 m across all surface types and depth magnitudes. The scatter plot demonstrates strong agreement with the 1:1 line across the full dynamic range from shallow urban ponding to deep channel inundation exceeding 12 m, with the $\log_{10}$ frequency colorbar confirming that the highest prediction density is concentrated along the diagonal rather than dispersed in error space. The 95% residual error range is narrow and tightly centered at zero, indicating that the model is unbiased at the domain scale and shows no systematic over- or underprediction. When the analysis is restricted to Urban nodes (Fig. 7C), MAE decreases to 0.017 m and RMSE to 0.047 m — lower in absolute terms than the global statistics, reflecting the shallower characteristic depths of urban street-level inundation relative to riverine and wetland environments. Critically, however, the residual distribution for Urban nodes remains leptokurtic and centered at zero, confirming that the Random Forest model captures the nonlinear threshold dynamics of urban drainage without introducing systematic bias at the sub-catchment scale where linear models fail most severely.

Fig. 8-A presents a scatter plot of model accuracy R^2 for every computational node in the domain. A clear performance hierarchy emerges. The Random Forest model demonstrates superior

predictive performance across all surface types, with most points falling above the 1:1 line. The vertical distance between the points and the dashed 1:1 line represents the "non-linear advantage". RF projection of max. The depth in Fig. 8-A shows that RF yields a more accurate regression for 92% of all nodes. Also, nodes classified as River (brown points) and Auxiliary Ditches (purple points) cluster in the upper-right quadrant, indicating high accuracy ($R^2 > 0.9$) for both linear and non-linear models in Fig. 8-A. This suggests that flooding in main channels would be physically simpler, driven primarily by hydraulic head gradients that are well approximated by a linear relationship. The linearity was also well maintained in the fluvial flood model, which provides boundary conditions for the urban flood model shown in Section 4.2, Fluvial Flood Driver Analysis. In contrast, Urban (blue points) and Ditch nodes exhibit a high variance in linear performance (PLSR R^2 ranging from 0.1 to 0.6) but show dramatic improvements under Random Forest (RF R^2 often exceeding 0.8). This confirms that urban flooding—governed by complex pipe networks, inlet restrictions, and microtopography—is inherently non-linear and cannot be adequately captured by additive statistical models, unlike linear models (Bulti & Abebe, 2020; Guo et al., 2020). Fig. 8-B highlights differences per surface type. Complex infrastructure, such as ditches and reservoirs, exhibits the largest variance in linear model error (red box), likely due to threshold behavior, such as capacity exceedance. The Random Forest model (green box) significantly reduces this variance, effectively capturing these binary non-linear dynamics. Although urban nodes show lower absolute RMSE due to shallower depths, Table 1 confirms that the Random Forest model doubles the explanatory power for these areas (R^2 increasing from 0.31 to 0.64 in Table 1), proving machine learning is essential for resolving street-level flooding.

This precision is further confirmed by the residual analysis. The Random Forest error distribution (Fig. 8-D) is sharply leptokurtic and centered at zero, indicating a systematic reduction in bias relative to the broader linear distribution. Furthermore, the error-reduction scatter plot (Fig. 8-C) shows a positive trend in deep flood zones, indicating that the non-linear model is particularly effective at correcting underpredictions in extreme events where linear averages fail. In summary, while riverine flooding follows linear driver interactions as in Section 4.2, urban and drainage-driven flooding is fundamentally non-linear, requiring the robust capabilities of Random Forest for accurate risk assessment.

Precipitation importance in Fig. 9-A is concentrated in inland areas with greater hydraulic distance from the tidal boundary, where rainfall-runoff generation controls peak depth independently of coastal forcing. The bimodal histogram confirms that this dominance is spatially selective rather than uniform, consistent with flash-flooding dynamics in urban sub-catchments, where impervious cover amplifies the local rainfall-depth response.

Downstream importance in Fig. 9-B shows the inverse pattern, with values exceeding 0.8 concentrated along the main channel network and coastal margins directly connected to the tidal boundary. The strongly right-skewed histogram confirms that surge-driven tailwater elevation is the dominant driver for the largest number of nodes, physically consistent with the backwater propagation mechanism identified in Section 4.2. Upstream importance remains negligible across the domain (Fig. 9-C), consistent with Section 4.2. Together, the three panels define a spatial gradient transitioning from surge-controlled boundary-adjacent zones (linear) to precipitation-controlled inland zones (nonlinear).

The cross-validated PLSR results provide a qualitatively distinct and diagnostically critical finding. For AuxDitches, River, and Wetland surfaces, PLSR achieves positive R^2 values that converge toward RF performance at n=32, confirming that linear models are adequate for these hydraulically simpler types, where flooding is governed by monotonic boundary-condition gradients. In contrast, for Urban, Ditches, and Water Reservoir surfaces, PLSR yields negative cross-validated R^2 across the entire n range, with confidence bands entirely below zero at n=32 (Urban: ~-2.0; Ditches: ~-1.1; Water Reservoir: ~-1.8). Negative R^2 indicates that the linear model generalizes worse than a null mean predictor. It is not merely less accurate than RF but structurally incapable of representing the flood response at these nodes regardless of ensemble size. This constitutes the strongest empirical evidence in this study for the non-linearity of urban flood physics: the failure of PLSR is not a sample-size artifact but an intrinsic property of threshold-governed hydraulic dynamics at urban and drainage-infrastructure nodes. We note that PLSR R^2 values in Table 1 reflect in-sample performance; the cross-validated learning curves reported here represent true generalization capacity and are the appropriate basis for the linear–nonlinear comparison. Additionally, RF for WaterReservoir plateaus at $R^2 \approx 0.62$, the lowest among all surface types, suggesting that discrete operational behaviors such as pump activation and gate management introduce a degree of deterministic variability that the three boundary condition drivers alone do not fully explain.

4.3.2 Residual Time Analysis

Flood residual time, defined as the duration for which surface water depth exceeds a critical threshold (5 cm in this study), is a key yet often-overlooked metric in flood risk assessment. Prolonged inundation significantly amplifies infrastructure damage through extended exposure to corrosive floodwaters, increases public health risks via waterborne disease transmission, and magnifies overall economic losses by extending business interruption periods (Cann et al., 2013; Dutta et al., 2003; Merz et al., 2010; Yang et al., 2025). Despite its importance, residual time presents unique analytical challenges that distinguish it from maximum water depth prediction. We applied PLSR as a baseline linear model alongside Random Forest (RF) and XGBoost, using the same framework as Section 4.3.1, but the results reveal a fundamentally more complex predictability structure.

The comparative model performance for residual time reveals a pattern that is structurally distinct from the maximum depth analysis. As shown in Fig. 10-A, the scatter of RF versus PLSR R^2 across all node clusters near the 1:1 line for urban, ditch, and inland surface types indicates that the performance advantage of the nonlinear model over the linear baseline is modest for these classes in contrast to the dramatic RF superiority observed for maximum depth. Fig.10-B confirms this: prediction error distributions for PLSR, RF, and XGBoost are broadly comparable across most surface categories, with overlapping interquartile ranges for Urban, Ditches, Wetland, and AuxDitches. RF reduces prediction errors relative to PLSR in 62.8% of nodes (Fig. 10-C); however, the magnitude of improvement is small, and the scatter is approximately symmetric around zero for most surface types, with the performance gap most pronounced at longer actual flood durations exceeding 25 hours. The residual error distributions (Fig. 10-D) for all three models are similarly concentrated near zero and have comparable spread, with XGBoost exhibiting only a marginally tighter peak — a modest contrast to the maximum depth analysis, where the RF error distribution was markedly more leptokurtic than the PLSR baseline.

Table 2 provides a quantitative performance summary that should be interpreted alongside the cross-validated learning curves shown in Fig. 12. For Urban and WaterReservoir, RF outperforms PLSR both in-sample (R^2 increasing from 0.57 to 0.72 and 0.41 to 0.61, respectively) and in cross-validation, consistent with nonlinear threshold dynamics governing duration at these nodes. Wetland shows strong performance across both models (PLSR: 0.86, RF: 0.88), confirming that flood duration in hydraulically connected tidal areas is approximately linearly predictable. Ditches show modest RF improvement (0.41 to 0.49). However, Table 2 shows a distinctive anomaly. For AuxDitches and River, RF R^2 is negative even in-sample (-0.05 and -0.20, respectively), indicating that the nonlinear model performs worse than PLSR for these surface types, regardless of whether training or test data are used. This could, in principle, reflect overfitting of the RF ensemble on 32 samples. However, the learning curves (Figure 12) do not support this interpretation: for these surface types, RF performance does not improve with increasing sample size and, in fact, degrades slightly, indicating that the three-scalar boundary-condition drivers simply do not contain sufficient information to explain residual time variability at these nodes.

The spatial distribution of RF feature importance for flood residual time at urban nodes (Fig. 11) reveals driver patterns consistent with those identified for maximum depth, thereby confirming the physical robustness of the importance attribution framework. Precipitation importance (Fig. 11-A) is concentrated in inland upland and mid-catchment areas where surface runoff generation governs flood response independently of tidal influence. The bimodal histogram — a dominant population near zero coexisting with a secondary population of high-importance nodes — confirms that precipitation dominance is spatially selective rather than uniform, reflecting the heterogeneous distribution of impervious cover and local drainage capacity across the domain.

Downstream importance (Fig. 11-B) is highest along the main channel network, coastal margins, and low-lying floodplain areas with direct hydraulic connectivity to the tidal boundary, where backwater propagation controls both the onset and recession of inundation. The right-skewed histogram with a dominant peak near 1.0 confirms that downstream surge height is the single most influential driver for the largest number of nodes — consistent with the PDP results discussed below. Upstream importance remains negligible for residual time (Fig. 11-C), consistent with the depth analysis.

4.3.3 Model convergence diagnostics

The preceding sections analyzed maximum water depth (Section 4.3.1) and flood residual time (Section 4.3.2) independently. This section directly compares the two response variables through model convergence behavior. This comparison reveals that depth and duration are governed by fundamentally different regimes of predictability — a distinction with direct implications for which modeling frameworks and input data requirements are appropriate for each component of compound flood risk assessment.

Figure 12 presents cross-validated learning curves for both response variables across six surface types, enabling a direct comparison of predictability scales with ensemble size and model complexity. For maximum depth (Fig. 12-A), two convergence regimes emerge. At surface types directly connected to the tidal boundary, including River, AuxDitches, and Wetland, all three models achieve positive R^2 by $n \approx 15$ and converge toward $R^2 > 0.85$ at $n = 32$. The adequacy of PLSR at these nodes confirms that flooding is governed by monotonic boundary condition gradients that can be represented linearly. At infrastructure-governed surface types, such as Urban, Ditches, and WaterReservoir, the two non-linear models converge to $R^2 \approx 0.65$–0.80 by approximately $n = 20$, while PLSR yields persistently negative cross-validated R^2 across the entire sample range. This divergence confirms that the failure of linear models at these nodes is an intrinsic property of threshold-governed hydraulic dynamics, not a sample-size artifact.

For flood residual time (Fig. 12-B), a qualitatively more complex structure emerges with three distinct predictability regimes. First, Wetland surfaces exhibit convergence for both non-linear models and PLSR ($R^2 \approx 0.75$–0.90), confirming that flood duration in tidally connected areas is approximately linearly predictable from scalar boundary conditions. Second, Urban, Ditches, and Water Reservoir surfaces show RF convergence at positive R^2 values (0.3–0.7), while PLSR remains persistently negative, paralleling the max depth pattern but at systematically lower R^2 values and with wider confidence bands, reflecting greater inherent variability than the peak depth response. Third, River and AuxDitches surfaces exhibit the most instructive contrast between the two panels: these same nodes that are readily predictable for maximum depth ($R^2 > 0.85$ in Fig. 12-A) yield negative R^2 for all three model architectures in Figure 12-B, with no convergence trend even at $n = 32$. This constitutes structural unpredictability, indicating that the scalar boundary conditions do not contain sufficient information to explain residual time variability at these channel nodes regardless of model complexity or ensemble size. This result shows that non-linear models substantially outperform the linear baseline for depth prediction, while their advantage for residual time is metric- and surface-type-dependent.

4.3.4 Comparative Driver Interaction Analysis

Fig. 13 presents per-node-averaged partial dependence plots for both response variables, enabling direct comparison of marginal driver effects. The downstream and precipitation responses confirm the trends identified in the global PDP (Fig. 6), showing monotonic downstream control, a precipitation saturation threshold at approximately 1.0 times baseline, and negligible upstream influence on peak depth. Two features emerge in the per-node analysis that are not resolved in the global model. First, precipitation saturation is more complete in duration than in depth once the drainage network is fully surcharged. Additionally, rainfall contributes to max depth accumulation but recedes relatively rapidly, so inundation persistence is effectively decoupled from precipitation intensity beyond the surcharge threshold. Second, the upstream duration response is non-monotonic:

$$\frac{\partial \grave{f}_{\mathrm{dur}}}{\partial x_{\mathrm{up}}}\begin{cases} < 0 & \text{if } x_{\mathrm{up}} < \dot{x}_{\mathrm{up}} \\ > 0 & \text{if } x_{\mathrm{up}} > \dot{x}_{\mathrm{up}} \end{cases} \tag{5}$$

where $\dot{x} \approx 0.65$ is the wetland storage saturation threshold (Fig. 13, bottom right) and $\grave{f}_{dur}$ is the partial dependence of residual duration on upstream stage x_{up}. Duration decreases as upstream water stage increases toward $\tilde{x}_{\mathrm{up}}$, consistent with the progressive engagement of upstream wetland floodplains that provide storage and create a hydraulic gradient permitting drainage away from the urban domain. Beyond this threshold, wetland storage capacity is exceeded, and the drainage pathway is eliminated, thereby increasing duration. The non-monotonic upstream response suggests that preserving or restoring upstream wetlands could serve as a nature-based surge-attenuation strategy.

To test whether these marginal responses act independently or modulate each other, Fig 14 presents the Friedman H-statistic analysis (Friedman & Popescu, 2008) alongside two-dimensional partial dependence surfaces. For maximum water depth, all pairwise interactions at Urban nodes remain below the weak-interaction threshold ($H^2 < 0.05$; Fig. 14-C), and the 2D PDP contours remain approximately vertical (Fig. 14-A), confirming that downstream surge and precipitation act as separable drivers on peak inundation depth. This separability is conditional on the fixed temporal structure of the Hurricane Matthew baseline, showing that events with different surge–rainfall phasing may produce depth interactions not captured here. The approximately vertical contours in Fig. 14A reflect the discrete sample increments but confirm the physical separability of downstream and precipitation drivers for peak depth at Urban nodes, indicating each driver shifts depth independently without modulating the other's marginal effect.

For flood residual time, a distinct interaction structure emerges. At Urban nodes, every driver pair shows weak but systematic interactions (H^2 = 0.06–0.09), with duration-to-depth H^2 ratios exceeding 2 times for Downstream vs. Precipitation and reaching 25 times for Downstream vs. Upstream. The 2D PDP for residual time (Fig 14-B) exhibits curvature in the contours at elevated values of both drivers, consistent with simultaneous drainage suppression from tailwater and network surcharge. This asymmetry reflects a fundamental difference between flood generation and recession: peak depth is determined by the single mechanism that dominates at a given node, whereas duration integrates over the entire recession phase, during which all drivers simultaneously affect drainage recovery.

Within the urban flood model domain, the interaction structure intensifies along the drainage conveyance pathway. At auxiliary ditches and river channel nodes, the Downstream × Upstream H^2 reaches 0.52 and 0.57, respectively (Fig. 14D). This concentration of interaction at channel nodes reflects the convergence of two hydraulic influences transmitted into the urban domain from opposite directions. The downstream boundary condition propagates tidal backwater directly into the urban channel network from the south. The upstream boundary condition, located far north of the urban domain, modifies water stages within the fluvial model, which in turn governs the boundary conditions delivered to the urban model along its northern interface. When both boundary stages are simultaneously elevated, the channel reach is hydraulically locked between these two externally imposed high-stage boundaries, and inundation persists until both boundaries independently recede. This compound boundary-condition effect provides a direct mechanistic explanation for the structural unpredictability of residual time at the River and AuxDitch nodes identified in Section 4.3.3 (Fig. 12-B).

4.3.5 Comparative Model Performance Across Response Variables.

A consistent cross-variable performance reversal is observed between the two nonlinear model architectures: Random Forest achieves superior or equivalent cross-validated accuracy relative to XGBoost for flood residual time prediction, while XGBoost shows a marginal but systematic edge for maximum water depth. This reversal is not a statistical artifact of sample size or hyperparameter sensitivity; it reflects a fundamental difference in the physical character of each response variable and in the mathematical mechanism by which each model architecture processes that character.

Maximum flood depth is a smooth, monotonically responsive quantity across the evaluated driver space, and deeper nodes tend to respond proportionally and predictably to driver magnitudes, producing a relatively smooth response surface without sharp discontinuities. This structure favors XGBoost's sequential gradient-boosting mechanism, which progressively refines predictions by correcting residuals along smooth gradients, thereby achieving high accuracy in the tails of the depth distribution, where linear models systematically underpredict (Biau & Scornet, 2016; Chen & Guestrin, 2016; Lundberg et al., 2020). The sequential residual correction is most effective precisely when the response surface is well-behaved, and the model can build incrementally on prior iterations without being misled by discontinuous threshold effects.

Flood residual time, by contrast, is a threshold-dependent variable: inundation begins and ends at discrete tipping points governed by interactions among drivers rather than by their scalar magnitudes alone. The onset of inundation depends on whether a critical drainage surcharge threshold is exceeded; its persistence depends on whether downstream tailwater elevation remains above the drainage recovery threshold throughout the event recession. Random Forest's ensemble of independently grown trees is architecturally better suited to capturing these threshold and interaction effects. Each tree can independently partition the driver space at different locations without being constrained by the residual structure of prior iterations, allowing the ensemble to collectively span a wide range of threshold configurations. This architectural property is most consequential where the response surface exhibits sharp boundary transitions, thereby precisely identifying the physical regime that governs urban flood duration.

4.4 Analysis of FEMA Flood Zones Regarding Flood Driver Importances

Historically, FEMA Flood Insurance Rate Maps (FIRMs) have successfully guided municipal floodplain management by delineating areas susceptible to primary coastal or riverine flood mechanisms. However, these traditional designations possess critical methodological limitations when assessing complex urban flood risk. Most notably, they rely on univariate hazard simulations that isolate individual drivers, thereby structurally neglecting pluvial (precipitation-driven) vulnerabilities in designated moderate-to-low-risk areas (Zone X) (Bates, 2023; Wing et al., 2022). Furthermore, the underlying regulatory models frequently employ steady-state or non-dynamic hydraulic simplifications. While computationally efficient, these static assumptions fail to capture the non-linear, transient hydrodynamics of compound flood events, such as synergistic backwater effects from simultaneous heavy rainfall and elevated downstream tailwater. Consequently, there is a pressing need to transition from static, single-driver mapping toward dynamic, multi-driver analytical frameworks.

To address this gap and explicitly test spatial non-homogeneity in flood sensitivity, this study employs an interpretable XGBoost machine learning framework that natively captures the non-linear interactions and spatial regime shifts inherent in compound urban flooding. Figure 15 presents the spatial distribution of historically flooded structures and repetitive-loss properties overlaid on FEMA flood zone designations (Fig. 15-A), alongside the comparative flood-driver importance stratified by FEMA zone and surface type (Fig. 15-B).

The catchment is partitioned into Zone AE, i.e., the high-risk 1% annual chance floodplain with established base flood elevations, and Zone X, representing areas of moderate to minimal flood hazard. Notably, 66% of repetitive loss areas and 65% of previously flooded structures are located in FEMA Zone X, which is designated as having lower flood susceptibility than Zone AE from OpenFEMA data. While the historical loss records reflect decades of diverse storm types and cannot be causally attributed to any single mechanism, the spatial coincidence between high historical losses in Zone X and the precipitation-dominated importance regime identified by the XGBoost analysis is consistent with the hypothesis that pluvial flooding in these areas is systematically underrepresented in current regulatory mapping.

The driver importance analysis using the XGBoost model reveals a physically coherent, near-binary stratification between the two flood zones. Zone AE nodes exhibit median downstream importance exceeding 0.8, with a compact interquartile range across Urban and Ditch surface types, confirming that surge height is the overwhelmingly dominant flood driver within the designated high-risk floodplain, where direct hydraulic connectivity to tidal conditions governs the flood response. Zone X nodes show a near-perfect reversal toward precipitation dominance, with median precipitation importance reaching 0.6–0.8 particularly for Urban and Ditch categories, reflecting that areas outside the active floodplain respond primarily to local rainfall-runoff dynamics in the absence of effective surge propagation. The wide interquartile range observed for Ditch nodes in Zone X reflects the heterogeneous drainage capacity of auxiliary conveyance infrastructure, where local imperviousness and channel capacity interact to modulate the precipitation-flood relationship non-linearly. Upstream importance remains negligible across both zones (consistent with Sections 4.3.1–4.3.2).

The convergence between FEMA zone boundaries, originally delineated through hydraulic modeling, and the mechanistic driver partitioning independently identified through the machine-learning sensitivity analysis strengthens confidence in both the physical interpretability and the regulatory relevance of the results. From a flood risk management perspective, surge mitigation strategies such as tide gates, levees, and coastal buffers are most critical within Zone AE, while green infrastructure, detention storage, and stormwater management interventions would yield the greatest benefit in Zone X, where precipitation dominates.

Nonetheless, the transitional zone between Zone AE and Zone X, where neither downstream nor precipitation importance clearly dominates, and both exhibit wide interquartile ranges, represents precisely the spatial regime where compound flood events pose the greatest and most underestimated risk. Compound flooding, defined as the concurrent or sequentially dependent occurrence of storm surge, heavy precipitation, and elevated riverine discharge, has been increasingly recognized as disproportionately more damaging than any individual driver acting in isolation. The Ditch surface-type nodes exhibit the widest variability in importance across both zones, reflecting their hydraulic susceptibility to simultaneous tailwater elevation from downstream surge and overloading from precipitation-driven inflow, a dual-sided hydraulic constraint that single-driver analysis cannot structurally resolve with marginal importance scores. The present sensitivity framework, by independently varying each driver within its prescribed range, quantifies individual contributions but does not capture the synergistic amplification that occurs when multiple drivers co-occur at high intensities. Future work should therefore extend this analysis toward joint probability sampling of the driver co-occurrence space to explicitly quantify non-linear interaction terms, providing a more complete basis for flood risk communication under projected increases in compound-event frequency associated with sea-level rise and intensifying precipitation extremes.

5. Discussion

5.1 Comparison with Prior Work

The transitional region between these spatial regimes represents the area of highest compound flood risk, exposing the greatest limitation of current single-driver assessments. The findings of this study contribute to the compound flood study along three dimensions that distinguish it from directly comparable prior work. First, regarding the treatment of flood driver nonlinearity, recent applications of tree-based machine learning to flood risk, including flash flood susceptibility mapping using XGBoost and RF (Abedi et al., 2022; Ma et al., 2021) and morphological flood impact quantification using XGBoost-SHAP (Wang et al., 2023), have demonstrated the predictive superiority of nonlinear models over conventional approaches but have not systematically quantified the physical source or spatial extent of this superiority. By deploying PLSR as a deliberate linear baseline and using cross-validated learning curves to distinguish structural model failure from sample-size effects, this study advances beyond predictive benchmarking to provide mechanistic proof of nonlinearity: the failure of PLSR at urban and drainage nodes is shown to be an intrinsic property of threshold-governed hydraulic dynamics, not a modeling artifact. This distinction has practical implications for where nonlinear modeling capacity is truly necessary in compound flood assessment.

Second, with respect to compound flood driver interactions, Xu et al. (2022) examined the compound flood impacts of water level and rainfall during tropical cyclones in Shanghai and demonstrated that simultaneous co-occurrence of drivers produces substantially greater inundation than either driver acting alone. However, that analysis characterized compound effects at the event scale through statistical correlation and did not resolve the spatial heterogeneity of driver dominance within the urban domain or distinguish between linear and nonlinear response regimes at the node level. The present study extends this line of inquiry by mapping driver importance at every computational node and demonstrating that the dominance of surge versus precipitation shifts spatially along a coherent hydraulic gradient, from tidally connected coastal zones to inland urban sub-catchments, with the transitional region exhibiting the greatest amplification of compound hazard. Similarly, Hossain Anni et al. (2020) showed that urban flood sensitivity varies significantly with stormwater infrastructure configuration, a finding consistent with the nonlinear threshold behavior identified here at Ditch and WaterReservoir nodes, but without the explicit linear-nonlinear comparative framework that allows those thresholds to be isolated and attributed.

Third, flood residual time, as an analytical target, has received comparatively little attention compared with maximum inundation depth and extent. Prior damage assessment frameworks, including the widely cited models of Dutta et al. (2003) and Merz et al. (2010), recognize inundation duration as a critical determinant of economic loss and public health impact. Yang et al. (2025) further demonstrated that prolonged inundation significantly increases the risk of hospitalization at a multi-country scale. However, these studies treat duration as an input to damage functions rather than as a variable to be predicted and attributed to physical drivers. The present study is among the first to subject flood residual time to a rigorous, node-level sensitivity analysis using both linear and nonlinear frameworks, and in doing so identifies a previously uncharacterized third predictability regime, structural unpredictability at AuxDitch and River nodes arising from the absence of temporal forcing information, that defines a clear and physically grounded data requirement for future compound flood duration forecasting systems.

5.2 Temporal Forcing Requirements for Flood Duration

The structural unpredictability of residual time at fast-responding riverine and auxiliary-ditch nodes (identified as Regime 3 in Section 4.3.3) exposes a fundamental limitation in the current boundary-condition framework. While scalar Latin Hypercube Sampling (LHS) successfully parameterizes peak magnitudes, which is sufficient for predicting maximum inundation depth, it inherently omits the temporal recession trajectory of the flood event.

To accurately characterize and predict inundation duration at these fast-responding nodes, future implementations of the compound flood sensitivity framework must transition beyond peak-magnitude scalar inputs and explicitly incorporate two additional categories of temporal forcing data: 1) Precipitation (Storm) Duration, 2) Surge Duration as possible candidates.

Total precipitation amount alone cannot distinguish between a short-duration, high-intensity burst and a prolonged, moderate-intensity event of an equivalent total volume. These scenarios produce fundamentally different residual time signatures. The former generates a sharp surcharge pulse followed by rapid recovery, while the latter sustains network surcharge and prolongs surface ponding well beyond the storm peak. Surge Duration denotes the temporal extent of the downstream storm-surge event, encompassing both the rise and recession limbs of the coastal boundary hydrograph. Surge duration directly controls how long the drainage network's outfall capacity remains suppressed by elevated tailwater. A surge of a given peak magnitude but short duration permits substantially earlier drainage recovery than a surge of equivalent peak magnitude sustained over a full tidal cycle.

5.3 Implications for Flood Risk Management

The spatially differentiated driver-importance maps and predictability-regime analysis provide directly actionable guidance for flood resilience investment. In FEMA Zone AE and other tidally connected areas where downstream surge dominates, confirmed by median XGBoost importance exceeding 0.8 across Urban and Ditch surface types, structural surge mitigation measures represent the highest-leverage interventions. These include tide gates, levees, coastal buffer restoration, and flap-valve upgrades at stormwater outfalls, prioritized not only to reduce peak flood depth but specifically to shorten inundation duration. The 3:1 asymmetry between downstream and precipitation marginal contributions to flood residual time means that surge control has a disproportionate impact on the metric that most directly determines infrastructure damage accumulation and public health exposure.

In FEMA Zone X and inland urban areas where precipitation-driven nonlinear threshold dynamics govern flooding, green infrastructure, distributed detention storage, and targeted stormwater capacity upgrades represent the most cost-effective risk reduction pathway. The transitional zone between Zone AE and Zone X, where compound interaction effects are most pronounced and importance metrics exhibit the widest interquartile ranges, represents the domain of greatest underestimated flood risk and warrants prioritized compound flood monitoring and joint probability hazard analysis.

Finally, the finding that accurate prediction of flood residual time requires storm and surge durations as explicit forcing inputs has direct implications for operational compound-flood early-warning systems. Current systems typically characterize flood hazard by forecasting peak discharge or water level, sufficient for depth prediction in most hydraulic regimes, but systematically insufficient for duration prediction in channel and auxiliary-ditch environments. Integrating storm- and surge-duration forecasts into compound flood warning algorithms is a necessary enhancement if these systems are to provide actionable guidance on inundation persistence under a changing climate, with increasing compound-event frequency and accelerating sea-level rise along the U.S. Southeast Atlantic coast.

5.4 Limitations and Future Directions

This study identifies the following three limitations, including independent driver sampling, temporal boundary condition constraints, ensemble size, and single-event baseline:

1. Independent Driver Sampling. The LHS sensitivity framework varies each flood driver independently, enabling clean first-order attribution but structurally decorrelating the input space. During landfalling Atlantic hurricanes, peak storm surge and precipitation totals exhibit a moderate-to-high positive correlation, indicating that the current ensemble systematically underrepresents simultaneous extreme driver combinations. Driver importance rankings should be interpreted as marginal sensitivities under a decorrelated input space. Future work must transition toward joint probability sampling frameworks, such as copula-based multivariate sampling conditioned on the regional climatological dependence structure, to explicitly capture the synergistic amplification of co-occurring extremes, particularly in the transitional zone between Zone AE and Zone X.

2. Temporal Boundary Condition Constraints. As established in Section 5.2, the scalar LHS boundary conditions encode peak and mean driver magnitudes but not the temporal structure, such as duration and phasing, that governs flood recession at hydraulically fast-responding channel and auxiliary ditch nodes. Incorporating precipitation duration and surge duration as explicit independent variables into the sampling framework represents the highest-priority methodological extension to improve residual time predictability. Additionally, the conclusion that downstream surge and precipitation act as separable drivers for peak depth is conditional on Hurricane Matthew's fixed temporal structure (surge–rainfall phasing held constant across all 32 scenarios); events with different phasing may produce depth interactions not captured here.

3. Ensemble Size and Single-Event Baseline. The sensitivity analysis is based on 32 scenarios derived from a single baseline event (Hurricane Matthew, 2016). While rigorous learning-curve analysis confirms statistical sufficiency for most surface types, the modest ensemble size limits the characterization of extreme tail behavior. Restricting the baseline to a single hurricane-type event also limits generalizability to non-hurricane precipitation regimes such as frontal systems or mesoscale convective complexes. Future work employing larger, stochastically generated multi-event ensembles would improve response-surface resolution at the parameter-space margins and broaden the applicability of driver-importance findings.

4. Driver Interaction Structure. The Friedman H^2 analysis demonstrates that driver interactions are metric-dependent and spatially differentiated: negligible for peak depth at Urban nodes, weak but systematic for duration at Urban nodes, and strong at channel nodes (AuxDitch $H^2 \approx 0.52$, River

$H^2 \approx 0.57$ for Downstream × Upstream). These findings are based on a decorrelated LHS ensemble and therefore characterize interaction structure under independent driver variation rather than under the positively correlated co-occurrence conditions of real hurricanes. Copula-based joint probability sampling conditioned on regional climatological dependence structure is needed for definitive quantification of interaction magnitudes under realistic co-occurrence scenarios

6. Conclusion

This study presents a dual-modeling framework that couples linear PLSR with non-linear Random Forest and XGBoost surrogates, trained on a 32-scenario LHS ensemble from an integrated HEC-RAS/PCSWMM hydrodynamic system, to quantify compound interactions among flood drivers in Savannah, Georgia. Three principal findings emerge.

First, compound urban flooding is fundamentally non-linear. Cross-validated learning curves confirm that PLSR yields persistently negative generalization R^2 for Urban, Ditch, and WaterReservoir surface types regardless of ensemble size, while Random Forest achieves R^2 exceeding 0.64 at these same nodes, demonstrating that the failure of linear models is an intrinsic property of threshold-governed hydraulic dynamics, not a sample-size artifact.

Second, flood driver dominance is spatially stratified along a coherent hydraulic gradient. Downstream surge controls flooding in FEMA Zone AE (median importance > 0.8), while non-linear precipitation thresholds dominate Zone X, where 66% of repetitive loss areas are located, exposing a systematic blind spot in single-driver regulatory mapping.

Third, flood residual time reveals a previously uncharacterized predictability regime. Channel and auxiliary ditch nodes are structurally unpredictable with any model architecture because scalar boundary conditions lack the temporal forcing information such as storm and surge durations required to resolve recession dynamics.

Methodologically, this work contributes (1) a node-level linearity gap diagnostic for determining where non-linear modeling is structurally necessary, (2) a learning curve protocol for sample-adequacy assessment in ML-based flood sensitivity analysis, and (3) a three-regime predictability typology distinguishing linear, non-linear, and structurally constrained flood response. Together, these contributions provide a transferable diagnostic framework for identifying where and why non-linear modeling capacity is structurally necessary, and for defining the data requirements that must be met before flood duration can be reliably forecasted in compound coastal environments.

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Table 1. Statistical performance summary comparing Partial Least Squares Regression (PLSR) and Random Forest (RF) across surface types. Values represent R^2, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

Surface Type	R2 (PLSR)	R2 (RF)	MAE (PLSR)	MAE (RF)	RMSE (PLSR)	RMSE (RF)
AuxDitches	0.97 ± 0.0	0.98 ± 0.00	0.031 ± 0.0	0.028 ± 0.00	0.041 ± 0.0	0.036 ± 0.0
Ditches	0.46 ± 0.3	0.70 ± 0.20	0.063 ± 0.06	0.033 ± 0.03	0.082 ± 0.07	0.063 ± 0.06
River	0.96 ± 0.1	0.97 ± 0.06	0.039 ± 0.04	0.033 ± 0.03	0.050 ± 0.04	0.043 ± 0.04
Urban	0.31 ± 0.3	0.64 ± 0.18	0.015 ± 0.03	0.008 ± 0.02	0.019 ± 0.04	0.014 ± 0.03
WaterReservoir	0.40 ± 0.3	0.65 ± 0.20	0.051 ± 0.05	0.027 ± 0.03	0.068 ± 0.06	0.056 ± 0.05
Wetland	0.97 ± 0.0	0.98 ± 0.03	0.031 ± 0.0	0.027 ± 0.00	0.041 ± 0.00	0.034 ± 0.00
Global (All Nodes)	0.51 ± 0.4	0.74 ± 0.22	0.027 ± 0.04	0.018 ± 0.02	0.035 ± 0.05	0.028 ± 0.04

Table 2. Statistical performance summary comparing Partial Least Squares Regression (PLSR) and Random Forest (RF) across surface types. Values represent R^2, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

Surface Type	R2 (PLSR)	R2 (RF)	MAE (PLSR)	MAE (RF)	RMSE (PLSR)	RMSE (RF)
AuxDitches	0.22 ± 0.29	-0.05 ± 0.43	1.77 ± 0.55	1.68 ± 0.29	2.89 ± 0.44	3.16 ± 0.64
Ditches	0.41 ± 0.28	0.49 ± 0.31	4.97 ± 3.14	3.09 ± 1.77	6.45 ± 3.70	5.62 ± 2.94
River	0.12 ± 0.17	-0.20 ± 0.30	1.80 ± 1.17	1.81 ± 0.65	3.10 ± 1.23	3.44 ± 0.96
Urban	0.57 ± 0.31	0.72 ± 0.30	4.16 ± 3.30	2.18 ± 1.93	5.36 ± 3.99	4.03 ± 3.41
WaterReservoir	0.41 ± 0.29	0.61 ± 0.22	7.27 ± 5.25	3.31 ± 1.55	9.40 ± 5.92	7.06 ± 2.97
Wetland	0.86 ± 0.19	0.88 ± 0.26	2.04 ± 0.72	1.07 ± 0.49	2.55 ± 0.88	1.62 ± 0.98
Global (All Nodes)	0.44 ± 0.36	0.40 ± 0.53	3.18 ± 2.81	2.04 ± 1.50	4.35 ± 3.30	3.74 ± 2.67

Figures

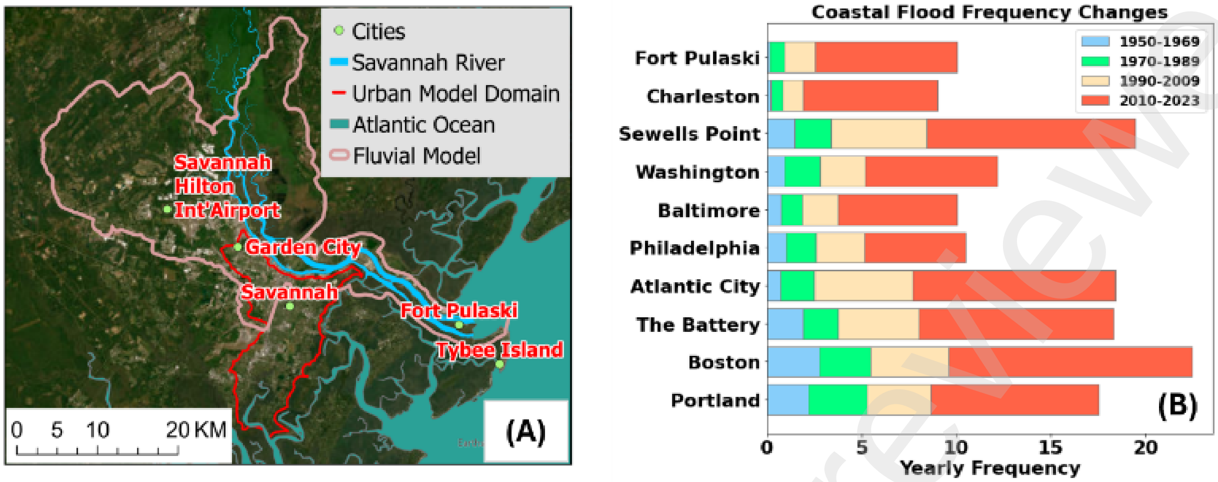

Figure 1. Research domain and rising patterns of coastal flooding around the Atlantic. (A) The research areas are shown with the fluvial flood model domain and the urban one; (B) The significantly increasing yearly frequency of coastal flooding at Fort Pulaski compared to major landmarks along the Atlantic (Data Source: NOAA Tide and Current).

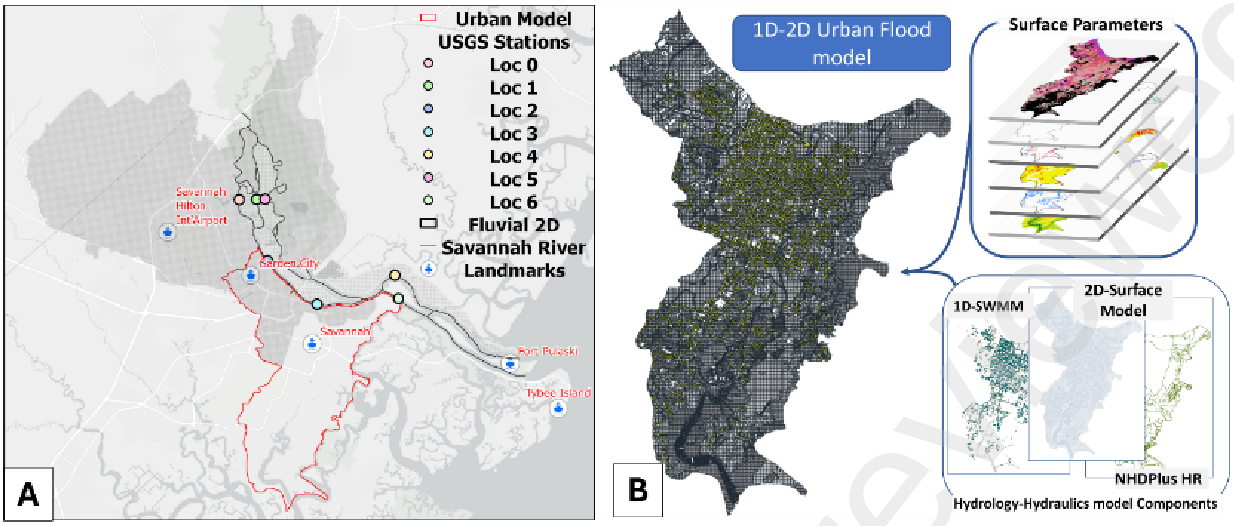

Figure 2. The integrated modeling framework includes a riverine flood model for the Savannah River Basin and an urban flood model for the City of Savannah. (A) 1D-2D HEC-RAS model domain and USGS monitoring stations; (B) 1D-2D urban flood model using SWMM/PCSWMM.

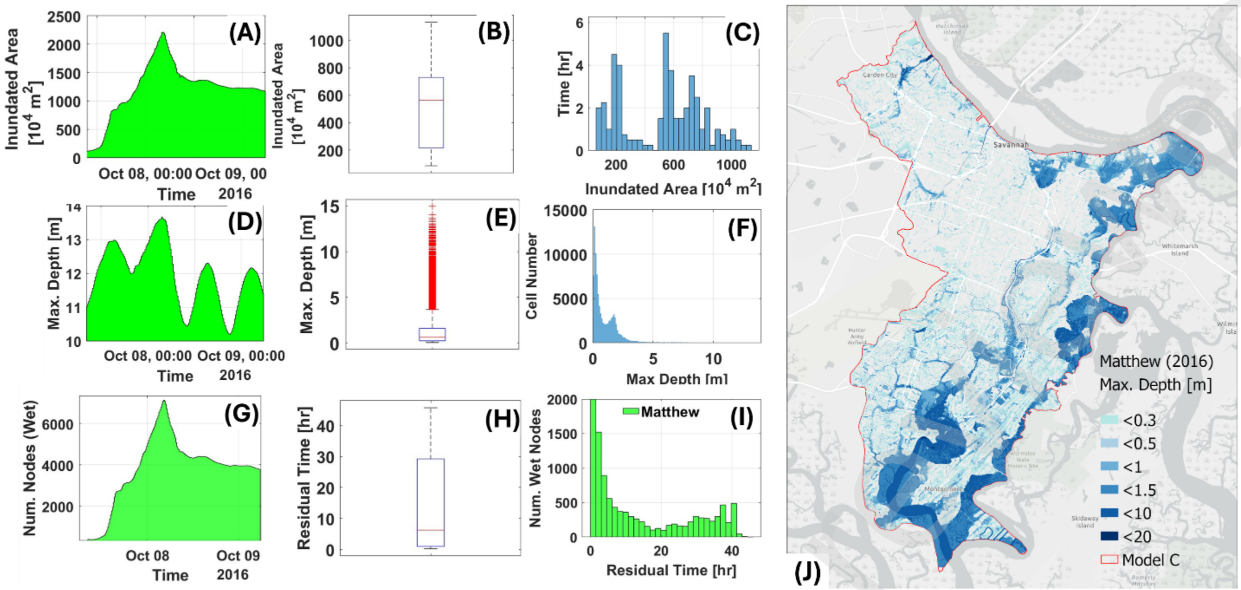

Figure 3. Baseline simulation results and empirical indicators of system non-linearity. (A–C) Inundated area evolution, statistical spread, and histogram of residual time. (D–F) Temporal changes, box plot, and histogram for maximum water depth. (G–I) Temporal changes, box plots, and histograms correlating residual time with the number of wet nodes. (J) Spatial distribution of maximum depth across the Savannah domain.

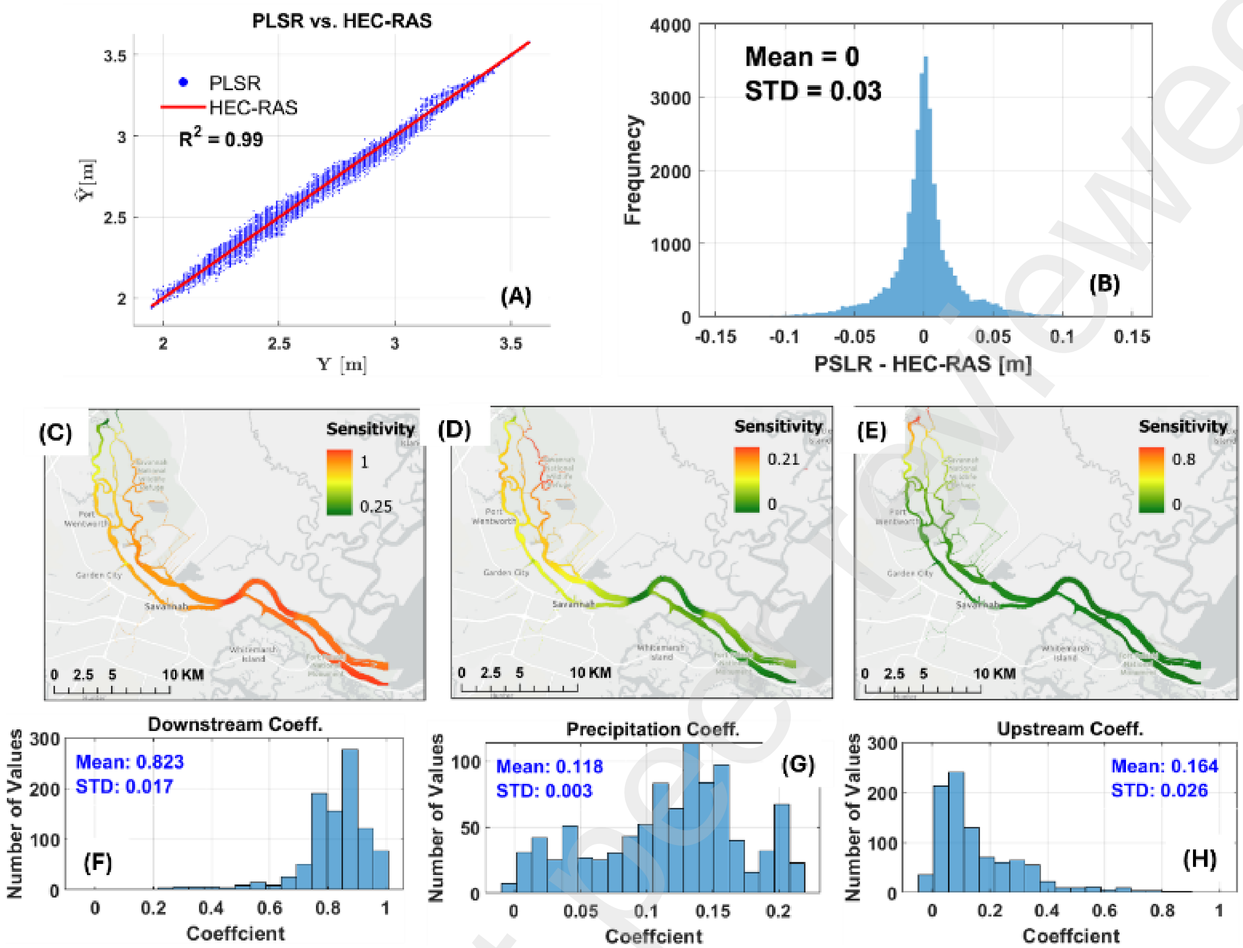

Figure 4. Linear boundary condition sensitivity analysis for the fluvial HEC-RAS model. (A) PLSR projection of MSWE compared to those from HEC-RAS; (B) Distribution of the difference between MSWE from PLSR and HEC-RAS; (C) Spatial distribution of PLSR weights related to downstream condition change; (D) Spatial distribution related to precipitation condition change; (E) Spatial distribution related to upstream condition change; (F) Histogram of the weight for downstream changes; (G) Histogram of the weight for precipitation changes; (H) Histogram of the weight of MSWE for upstream changes.

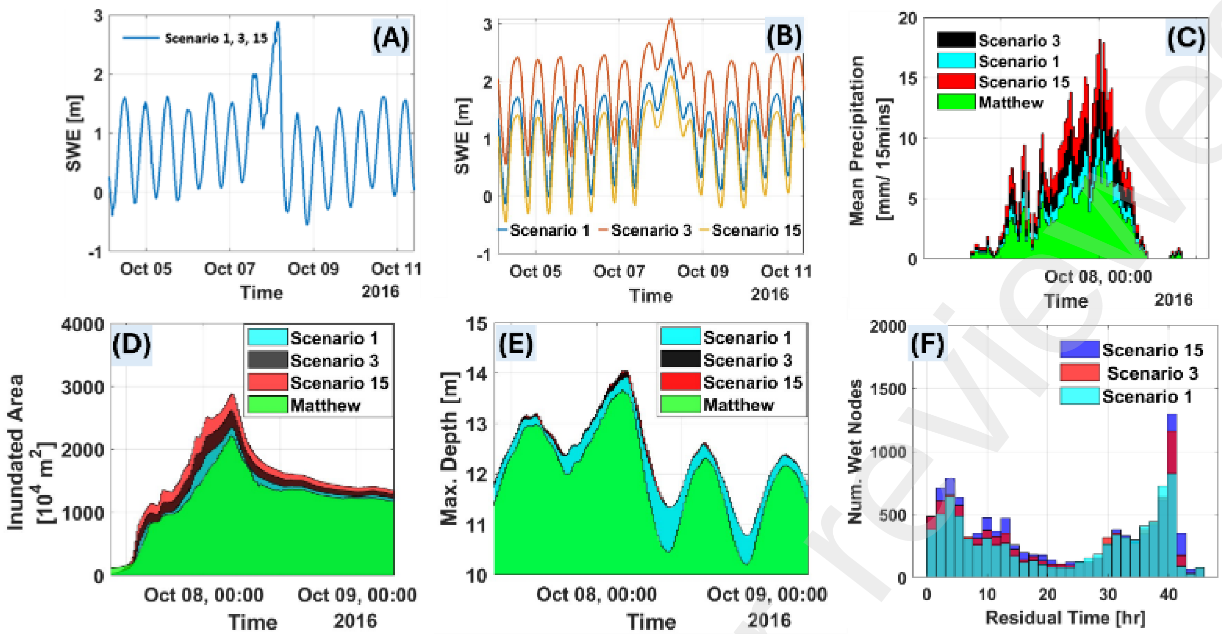

Figure 5. An example of simulations for specific scenarios showing inundated areas, maximum depth, and residual time per node. (A), (B), and (C) represent downstream condition, upstream condition, and precipitation conditions, respectively; (D) compares inundated areas across the scenarios; (E) shows maximum depth change over time; (F) displays the number of (Wet) nodes per residual time.

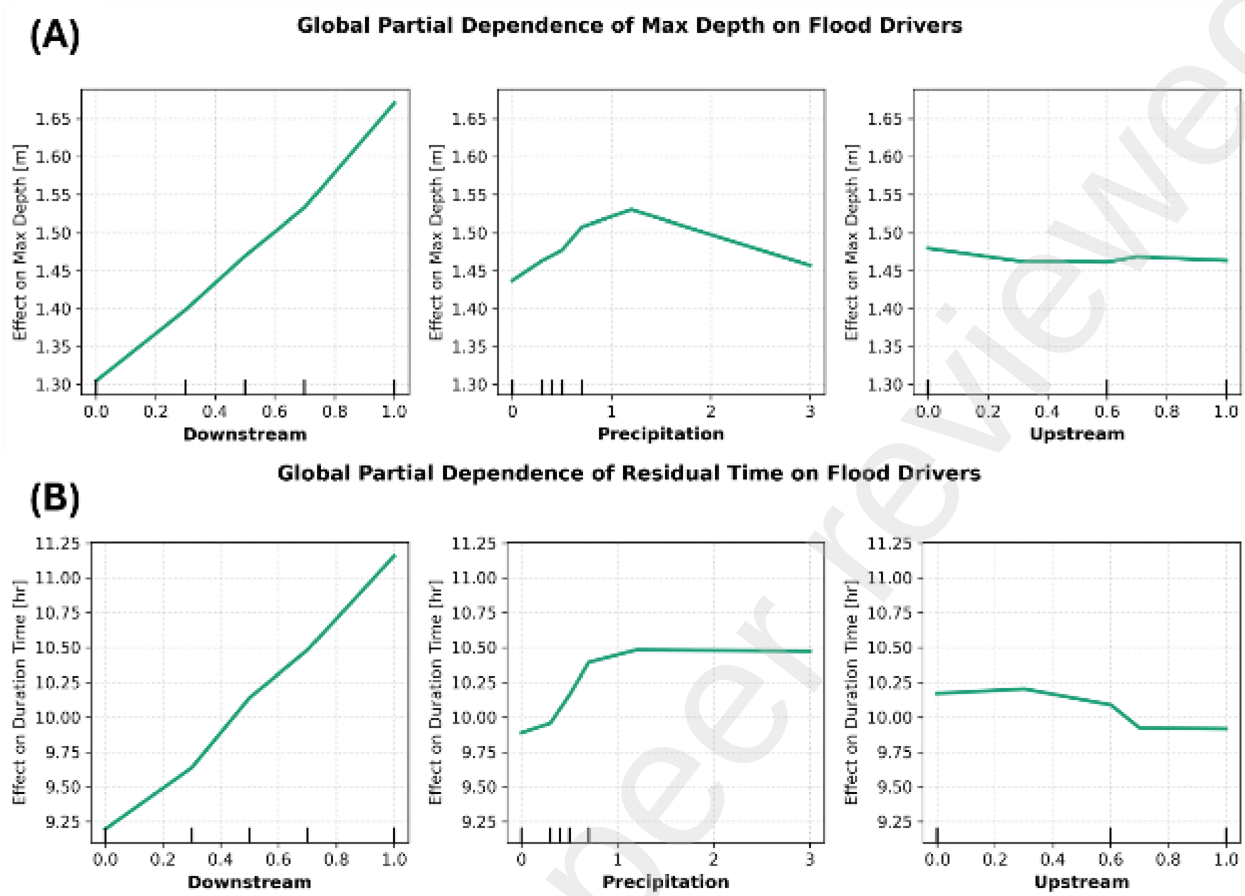

Figure 6. Global partial dependence of maximum flood depth and flood residual time on primary boundary conditions. (A) Maximum depth and (B) flood residual time response to downstream conditions, precipitation, and upstream inflow, derived from an XGBoost surrogate model trained on the full pooled dataset across all computational nodes within the urban flood model domain.

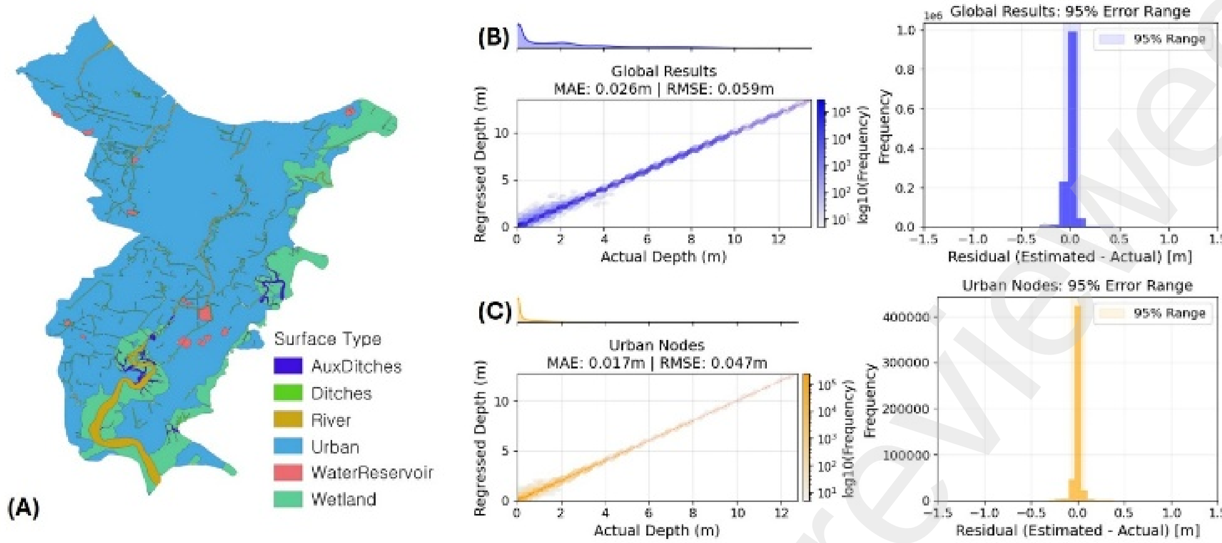

Figure 7. Random Forest (RF) predictive performance for maximum water depth. (A) Surface classification map across the urban flood model domain. (B) Domain-wide comparison of RF-regressed maximum depth versus simulated depth, including residual error distribution. (C) Predictive performance and residual error distribution isolated strictly to Urban nodes.

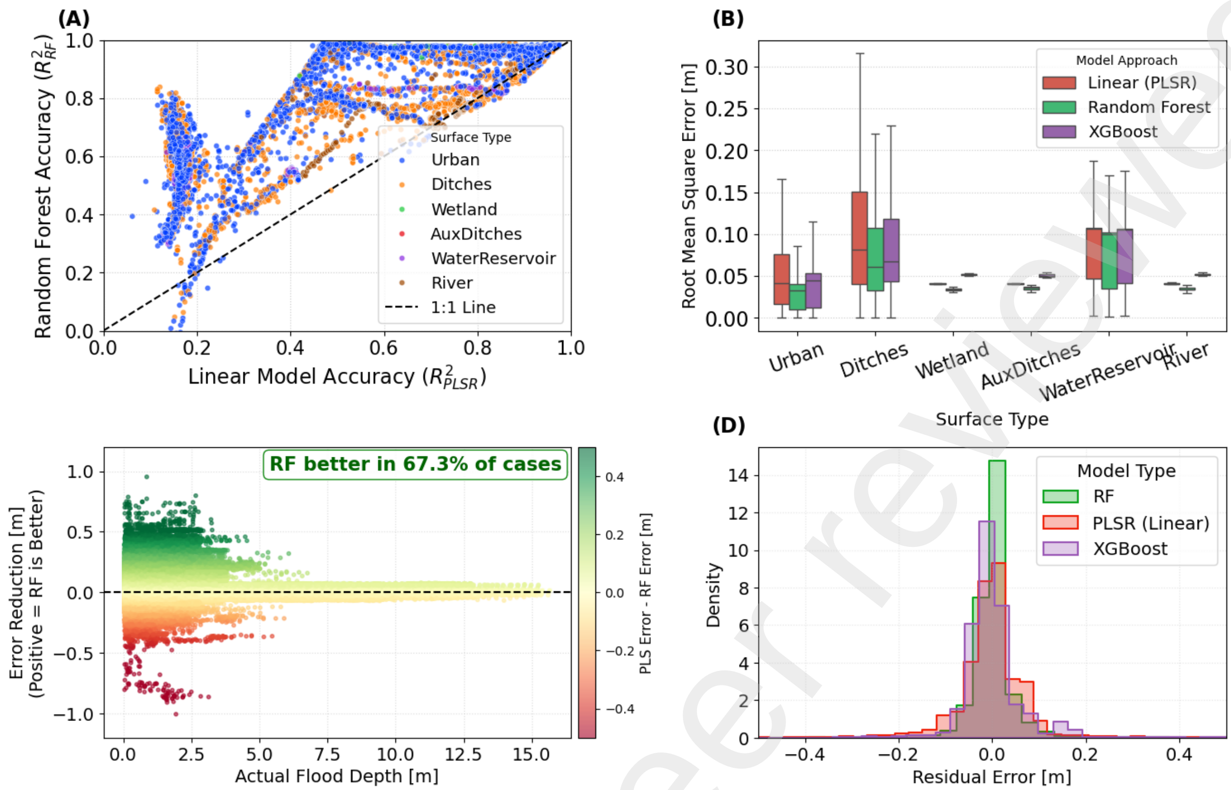

Figure 8. Comparative performance analysis of Linear (PLSR) versus Non-Linear (Random Forest) models. (A) Scatter plot of model accuracy (R^2) for all nodes, where points above the diagonal indicate superior Random Forest performance. (B) Boxplot of Root Mean Square Error (RMSE) distribution across different surface types, highlighting significant error reduction in Ditches and Urban areas. (C) Error reduction magnitude (PLS Error - RF Error) versus actual flood depth; positive values indicate instances where Random Forest provided a more accurate prediction. (D) Histograms of residual errors ($Depth_{Actual} - Depth_{Projected}$), showing tighter precision and reduced bias for RF and XGBoost.

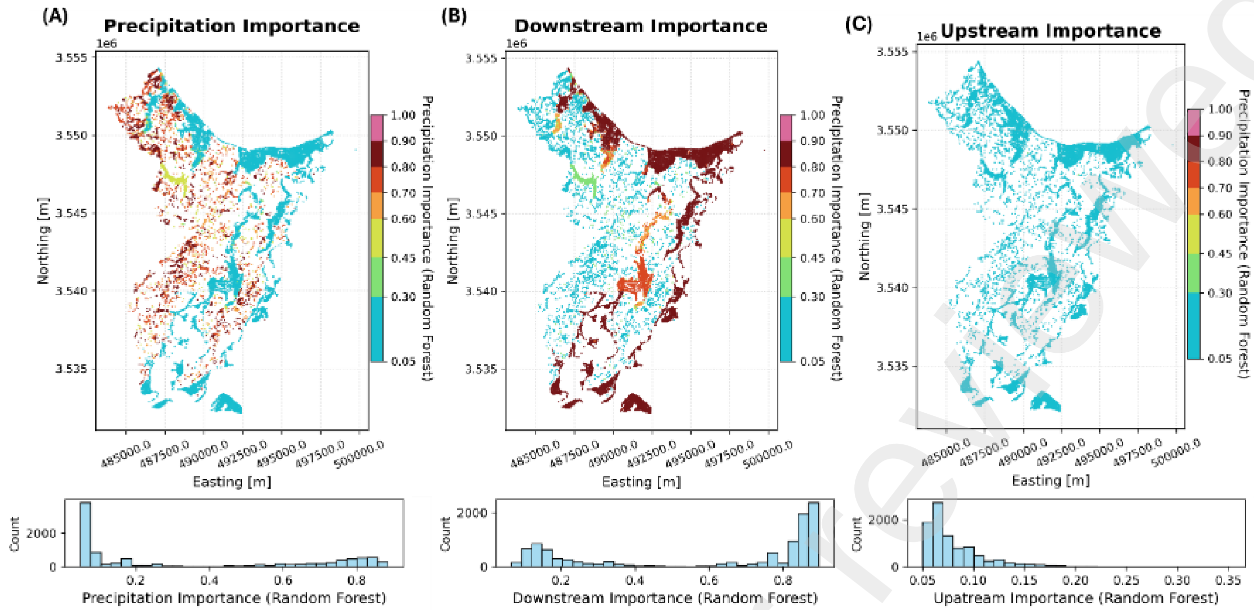

Figure 9. Spatial distribution of Random Forest feature importance for maximum water depth. (A) Precipitation importance, highlighting concentrated influence in inland sub-catchments. (B) Downstream (coastal) importance, demonstrating dominance along tidal networks. (C) Upstream importance, showing negligible influence domain-wide.

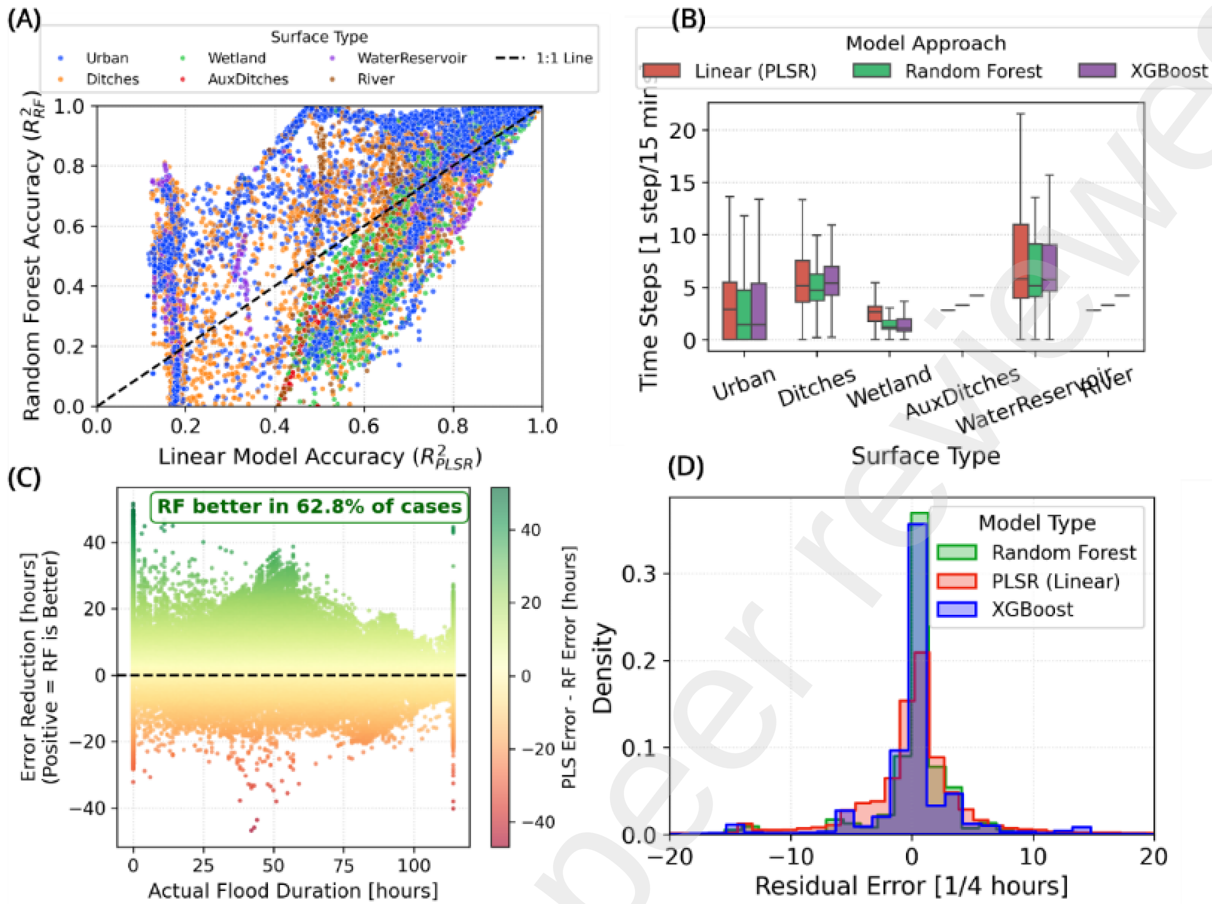

Figure 10. Model comparison based on land use type and types (PLSR, RF, and XGBoost). (A) Scatter plot of model accuracy (R^2) comparing Random Forest against the linear PLSR baseline. (B) Boxplot comparing prediction errors (in 15-minute time steps) across all three models, categorized by surface type. (C) Scatter plot of error reduction demonstrating RF's superiority over PLSR across varying actual flood durations, with RF performing better in 62.8% of cases. (D) Density histogram of residual errors, highlighting the tighter precision of the XGBoost model.

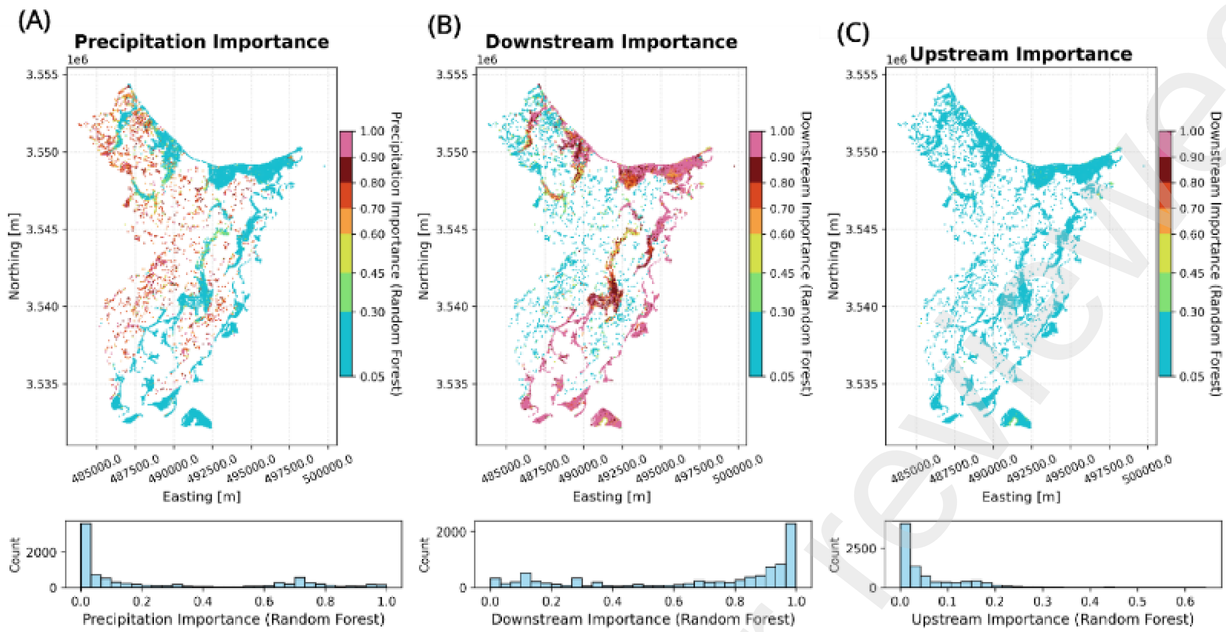

Figure 11. Spatial distribution of Random Forest feature importance (gain) for flood residual time prediction across areas assigned as Urban. (A) Precipitation importance, demonstrating localized regions of strong influence. (B) Downstream importance, showing domain-wide dominance and widespread high-importance values. (C) Upstream importance indicates a negligible physical impact on residual time across the vast majority of the study area.

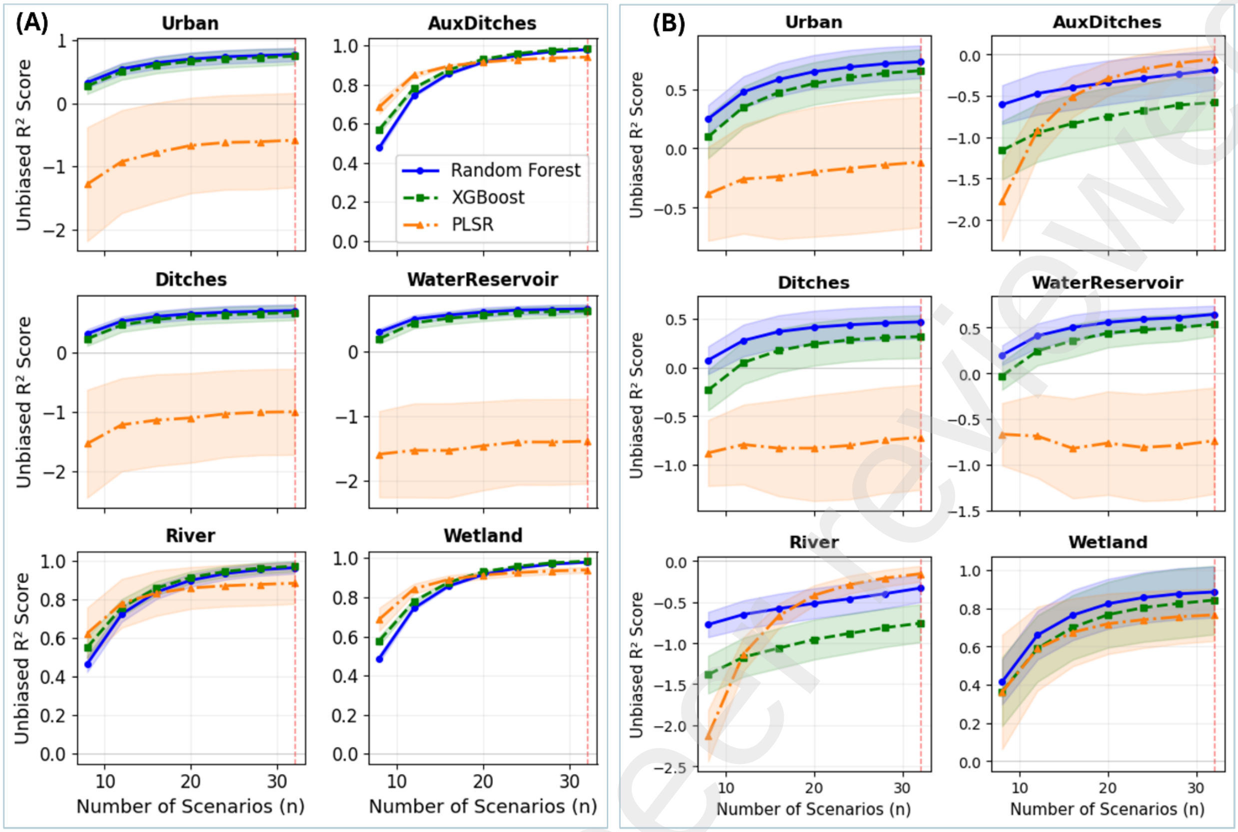

Figure 12. Cross-validated learning curves for maximum flood depth (A) and flood residual time (B) across six surface types. Maximum depth: Random Forest (blue, OOB), XGBoost (green, CV), and the linear PLSR baseline (orange, CV) as a function of training scenario sample size (n = 8 to 32). Shaded bands indicate ±1 standard deviation across 50 bootstrap resamplings.

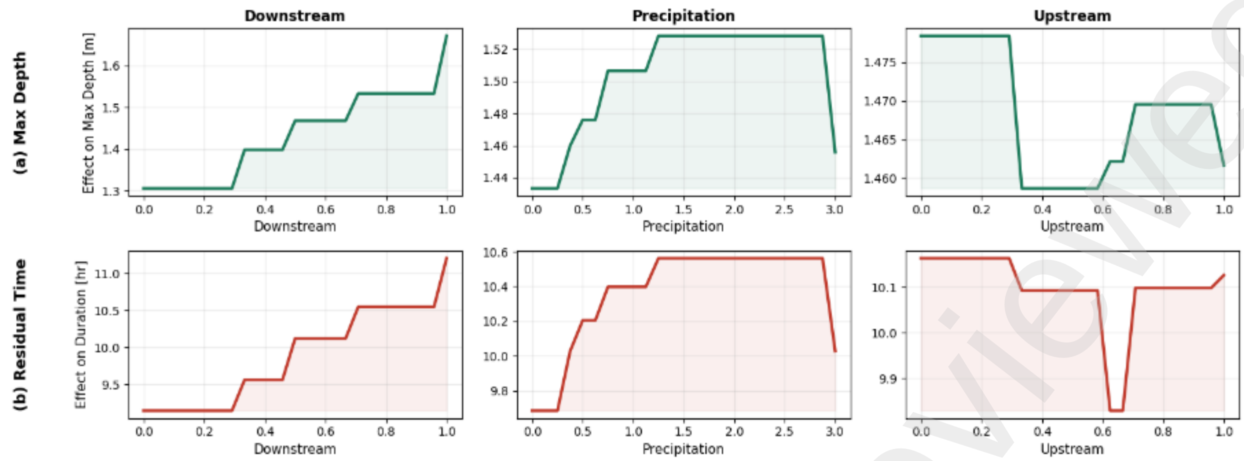

Figure 13. Per-node-averaged partial dependence on maximum flood depth. (a) and flood residual time, (b) on primary boundary conditions, derived from per-node XGBoost surrogate models across the urban flood model domain.

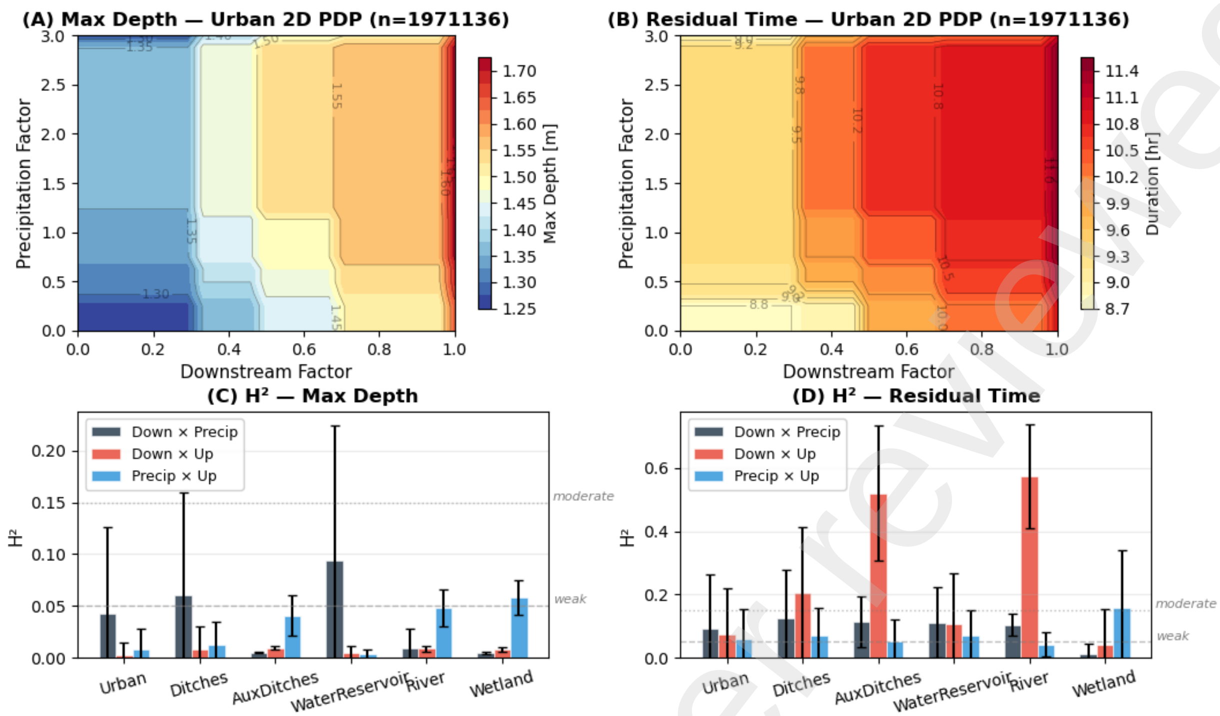

Figure 14. Driver interaction analysis comparing maximum flood depth and flood residual time. Two-dimensional partial dependence on Downstream × Precipitation at Urban nodes for maximum depth (A) and residual time (B). Friedman H² interaction statistic by surface type for maximum depth (C) and residual time (D). Dashed and dotted lines indicate weak (0.05) and moderate (0.15) interaction thresholds.

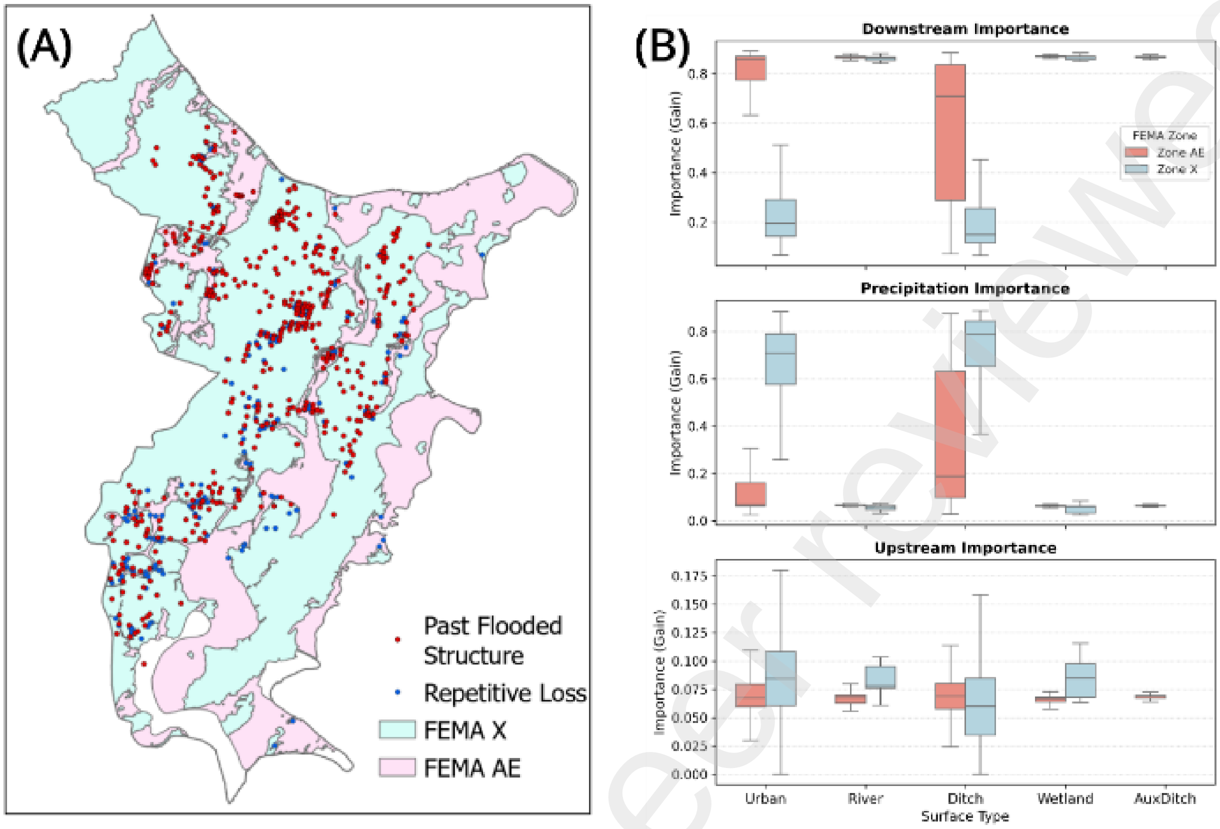

Figure 15. Spatial distribution of historically flooded structures and repetitive loss properties according to flood zone and importance. A) Spatial distribution of historically flooded structures and repetitive loss properties overlaid on FEMA flood zone designations. (B) Flood driver importance stratified by FEMA zone and surface type.